



Overview

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Official Course Outline

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Course Outline (Syllabus)

3. Proposal: 3-Minute Talk (3MT)



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Settings

Add dates and restrictions...

Add a description...

Upload / Create

Existing Activities

Bulk Edit

3 Project Presentation: 3-Minute Talk (3MT)



Assignment

Due October 15 at 5:00 AM

Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

Don't be squared ..., Pulp Fiction, Quintine Tarantino, 1994

<https://www.youtube.com/watch?v=IY48IXSK4nk>



Logical Level | Relational Model

3

A Data Odyssey: Relational DBMS vs. GGG



Data Rep.
in
RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model Level

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep.
in
GGG

Real World Entity

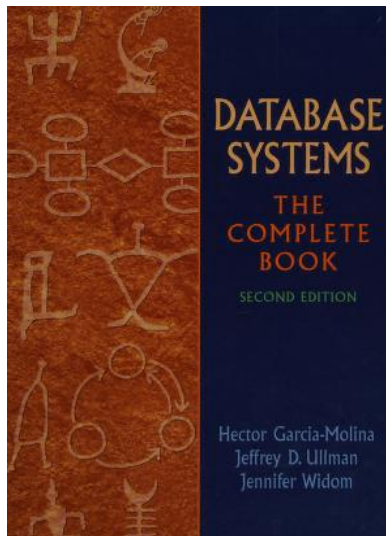
Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

Outline: W04A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation

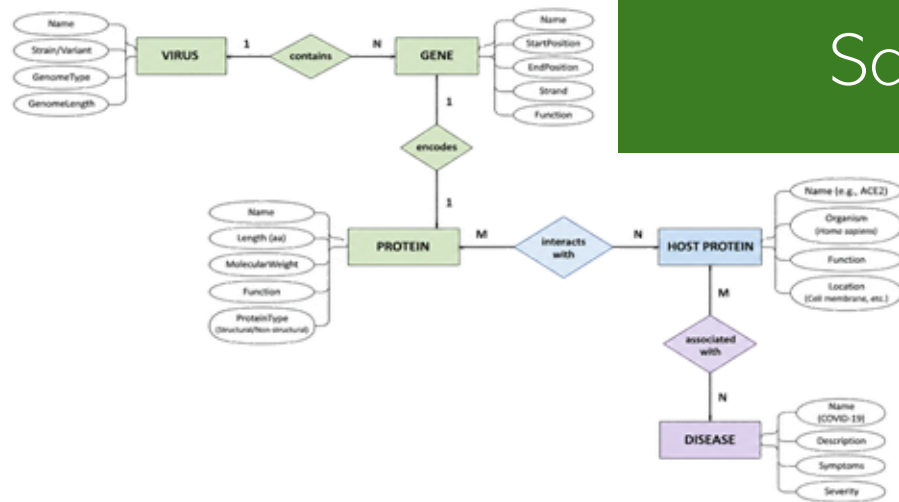
IIII) IS-A to Relation

Hands-On

5

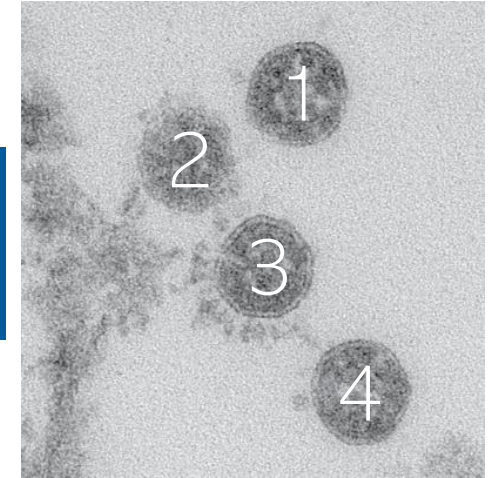
Abstractions: Logical Representation

How **instances** of entities, **values** of their attributes, their relationships should be represented based on a schema.



Schema

Real-World
Instances of Entity



Storage

Abstractions: Logical Representation

Common logical models include:

~1960: Object Oriented

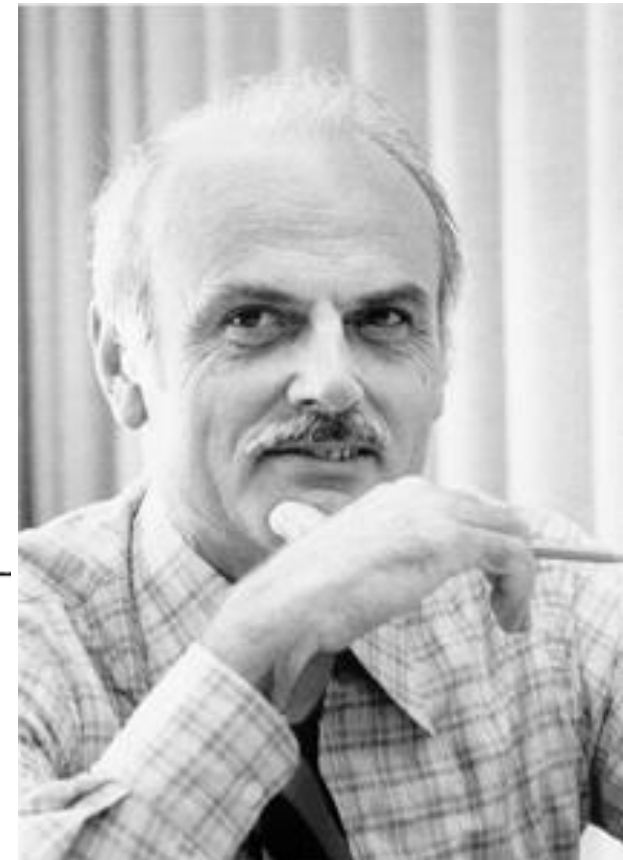
~1969: Relational: Mathematical Relation

~1996: XML

Edgar Frank "Ted" Codd, IBM, 1969, 1970
Information Retrieval

**A Relational Model of Data for
Large Shared Data Banks**

E. F. CODD
IBM Research Laboratory, San Jose, California



7 Logical Model: Relational

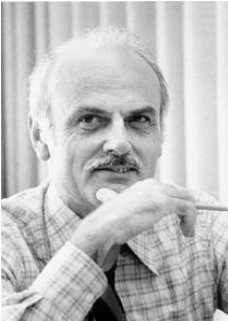
An Entity Set with Attributes coming from domains $A, B, C, \dots$, is like,

$$A \times B \times C \times \dots = \{(a, b, c, \dots) \mid a \in A, b \in B, c \in C, \dots\}$$

that can include all possible entities of the Entity Set.

→ Data instance is then represented as a **Tuple**: $(a, b, c, \dots)$

→ Entity Set is then a **Relation**: $\{(a, b, c, \dots), (a', b', c', \dots), \dots\}$



Relational Model: Formal vs. Informal Terminology

Relation

Schema	<u>Title</u>	ReleaseDate	RunningTime	Genre
Tuple 1	2001: A Space Odyssey	English	1 42	SciFi
Tuple 2	Rosemary's Baby	English	1 36	Horror
Tuple 3	The Birds	English	1 19	Horror
...	...	...	...	

Relation vs. Table

Tuple vs. Row

Domain vs. Column

Relational Model: Everything is Relation

Entity Set $\rightarrow$ Relation

Relationship Set $\rightarrow$ Relation

Data Modeling
in
RDBMS

Real World Entity

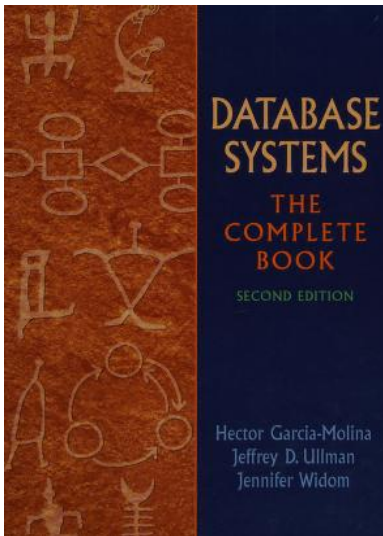
Conceptual Level | Entity Set & Relationship Set

Conceptual Level | Logical Level | Relation

Conceptual Level | Logical Level | Physical Level | SQL

Conceptual Level | Logical Level | Computable Entity

Outline: W04A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

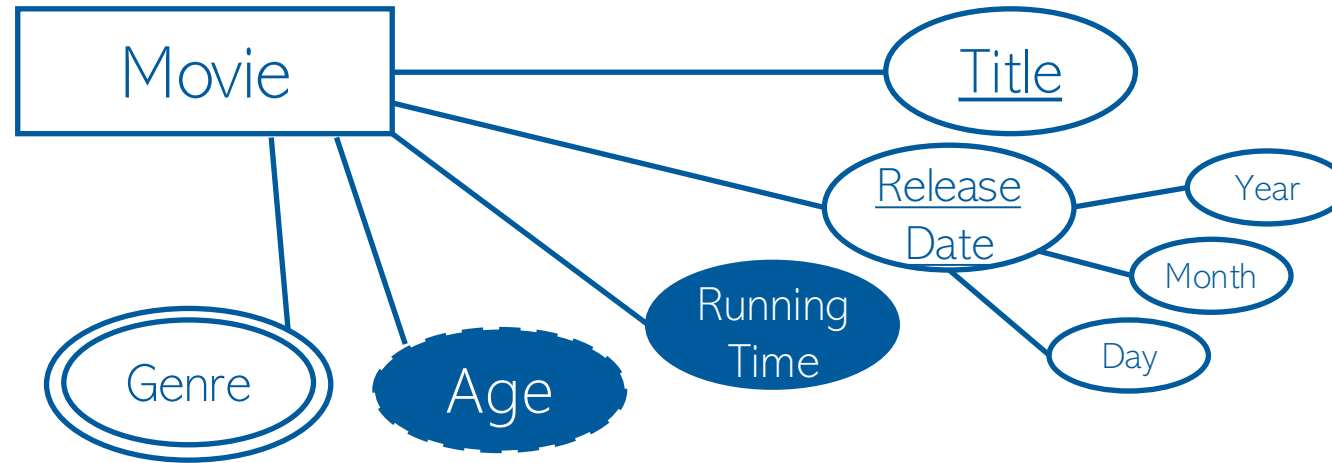
III) Relationship to Relation

IIII) IS-A to Relation

Hands-On

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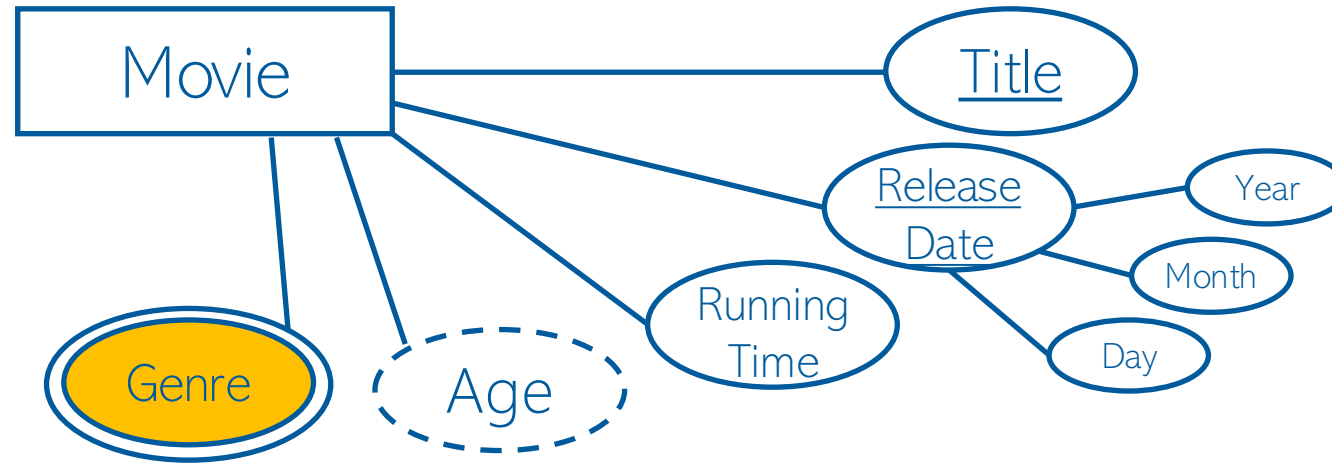
Relational Model: ER to Relation Transformation



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

12

Relational Model: ER to Relation Transformation

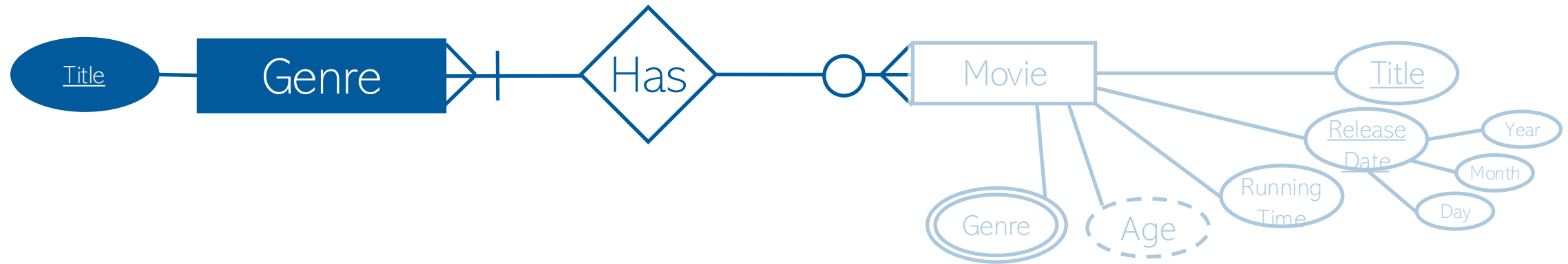


Relational model does not have multivalued attributes!

R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

13

Relational Model: ER to Relation Transformation

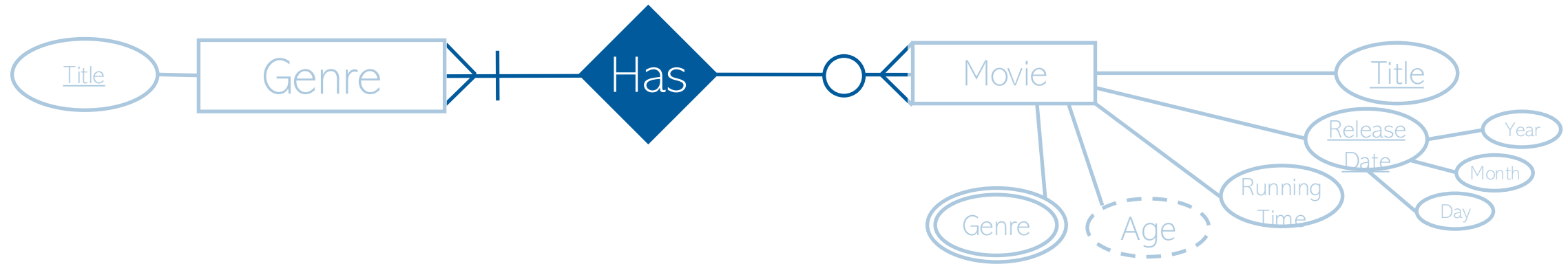


R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Genre(Title)

14

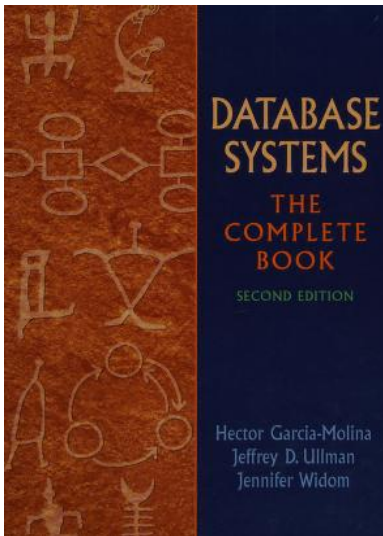
Relational Model: ER to Relation Transformation



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Genre(Title)

Outline: W04B



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

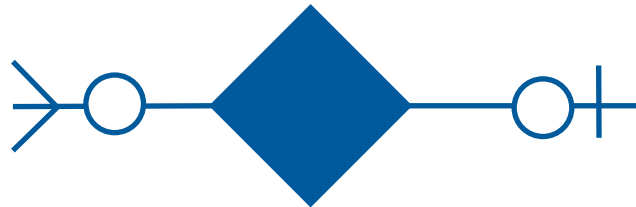
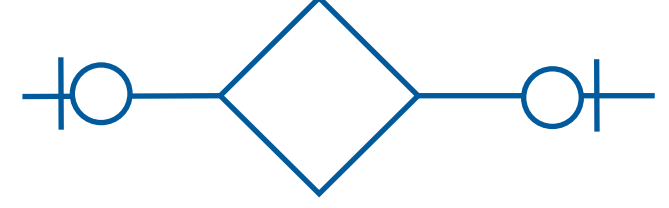
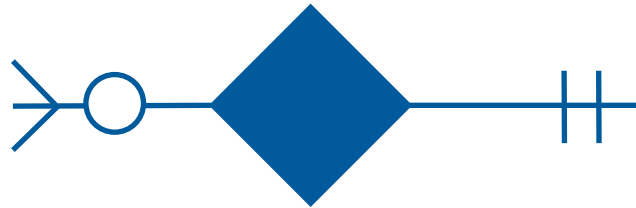
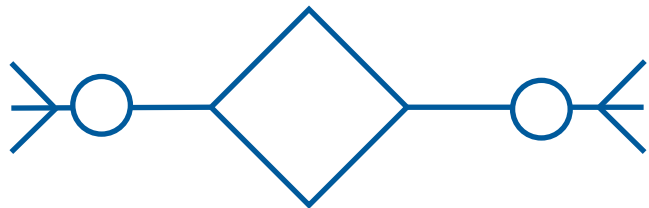
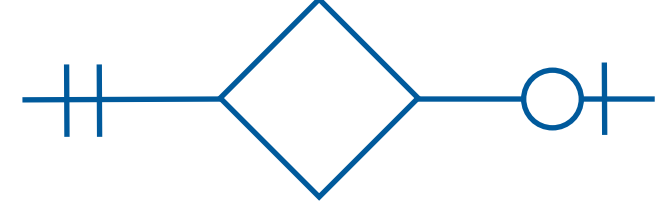
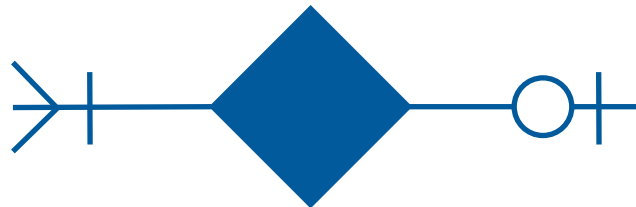
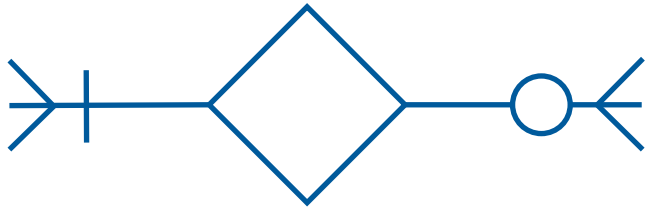
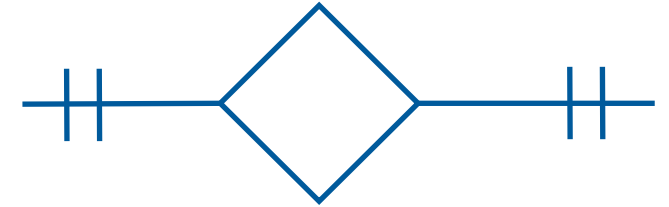
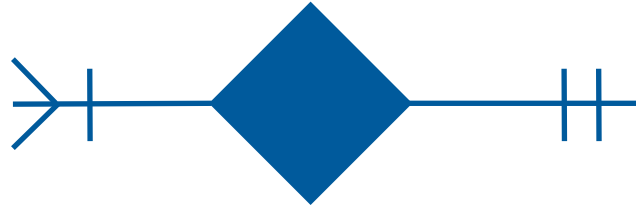
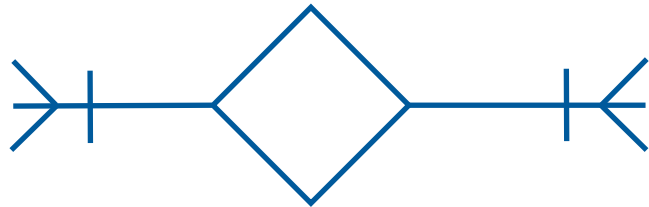
III) Relationship to Relation (Many-One)

IIII) IS-A to Relation

Hands-On

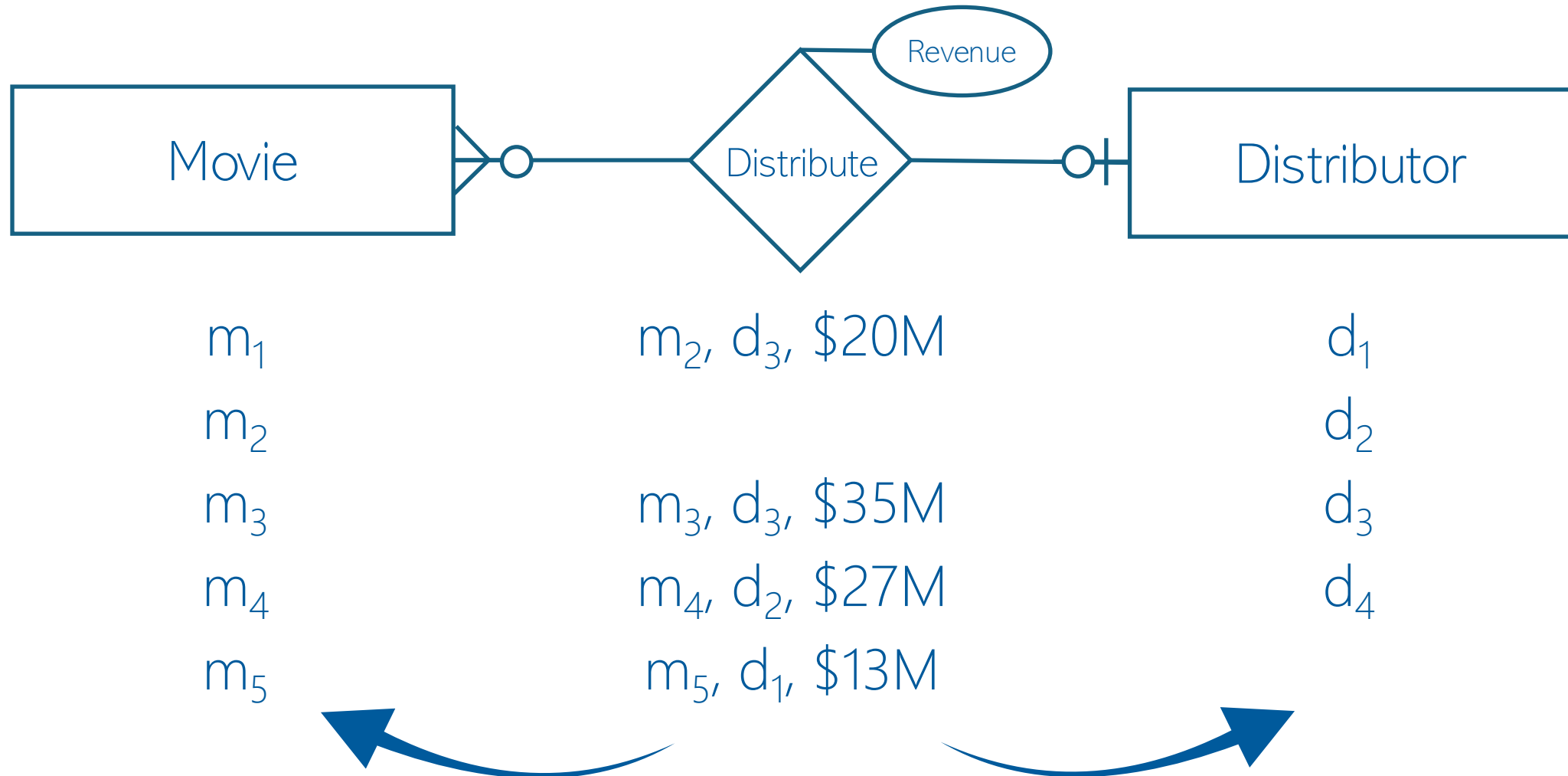
16

Relationship: Many-One (One-Many)



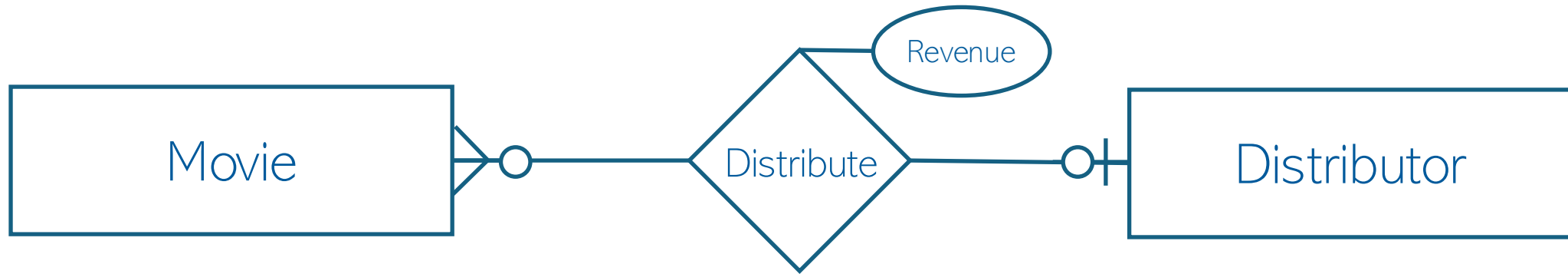
17

Relationship: Many-One (One-Many)



18

Relationship: Many-One (One-Many)

 m_1 m_2 m_3 m_4 m_5

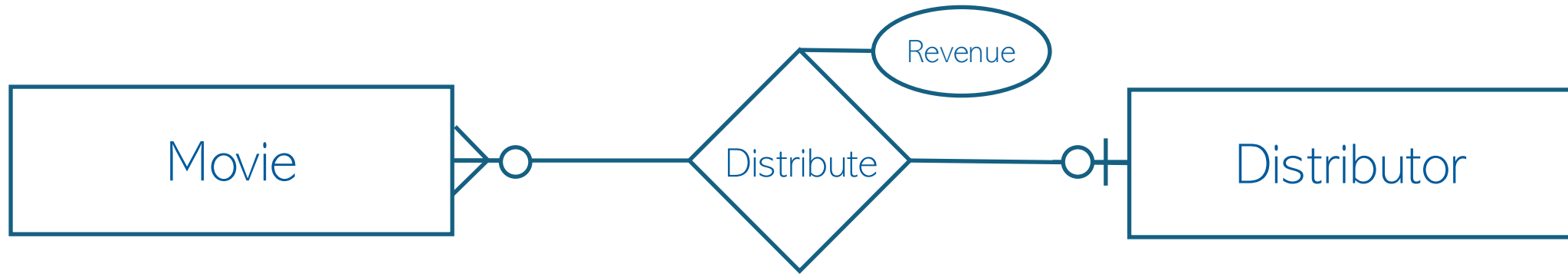
A distributor needs new attributes
for a new movie to distribute!

Inserting data instance needs
changes to schema!

 $d_1, m_5, \$13M$ $d_2, m_4, \$27M$ $d_3, m_2, \$20M, m_3, \$35M$ d_4 

19

Relationship: Many-One (One-Many)



NULL, NULL, m_1

d_3 , \$20M, m_2

d_3 , \$35M, m_3

d_2 , \$27M, m_4

d_1 , \$13M, m_5

As a movie can have maximum 1 distributor, if any, the number of attributes to add is always fixed.

d_1

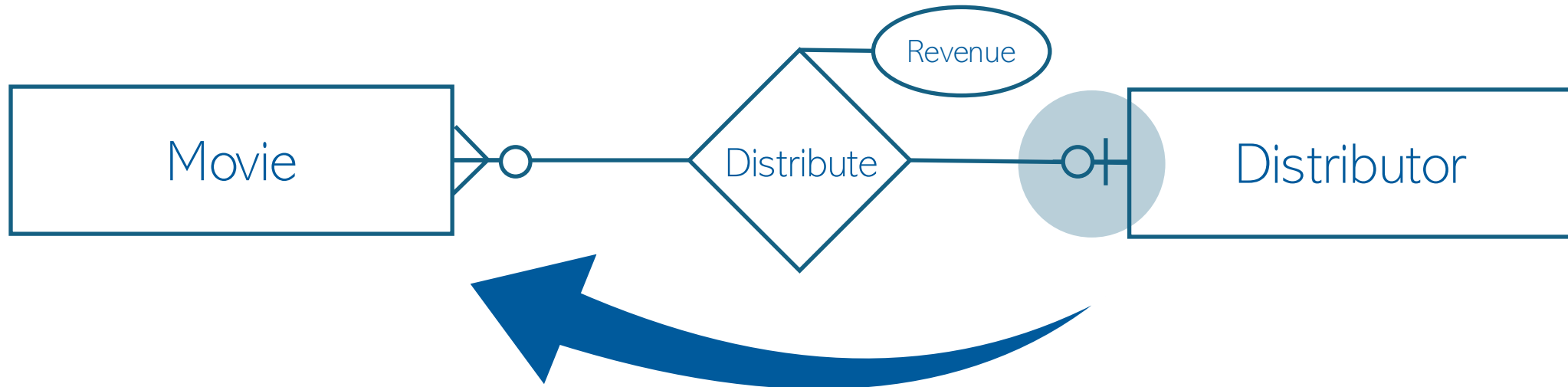
d_2

d_3

d_4

20

Relationship: Many-One (One-Many)

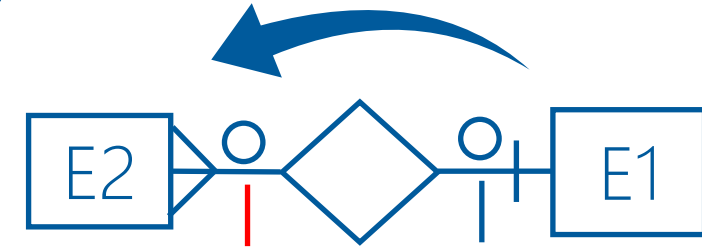


Everything goes to entity set with cardinality one (i.e., visually many side)
Because it only needs to store one relationship at max.

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Relationship: Many-One (One-Many)

Input: Many-One relationship between E2 and E1, i.e.,
Output: Relations R1 for E1 and R2 for E2.

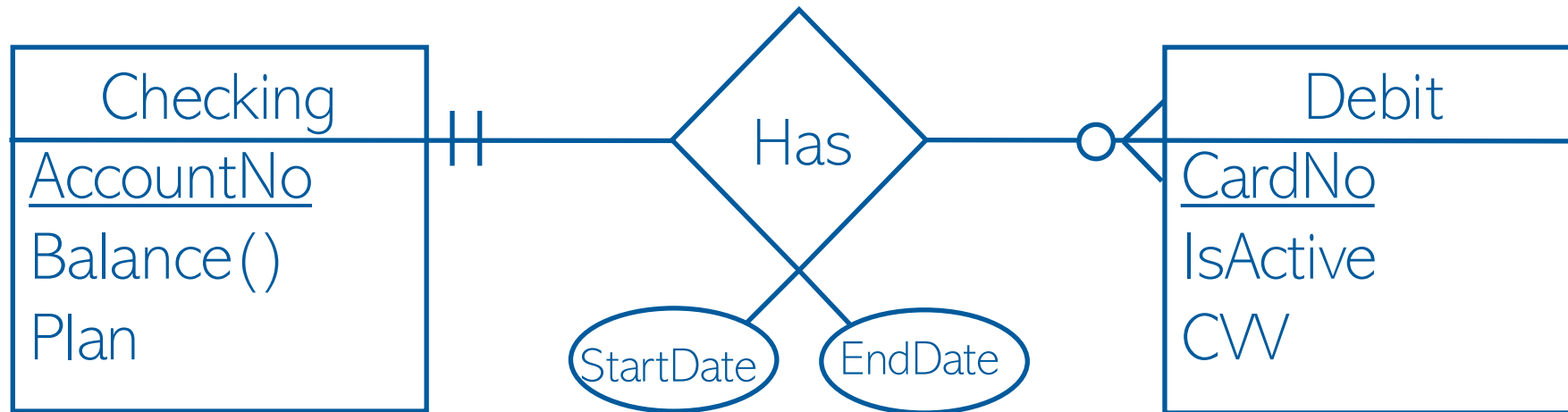
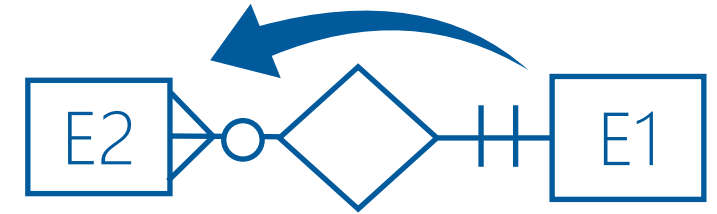


1. For E1, create relation R1 with the same attributes and keys as in E1
2. For E2, create relation R2 with the same attributes and keys as in E2
3. **[Foreign Key Set]** Add key set of E1 to R2
4. Add attributes of relationship set to R2
5. If E2 ordinality is optional then make foreign key set optional
else make foreign key set mandatory

Herein, we do not care about E1's ordinality.

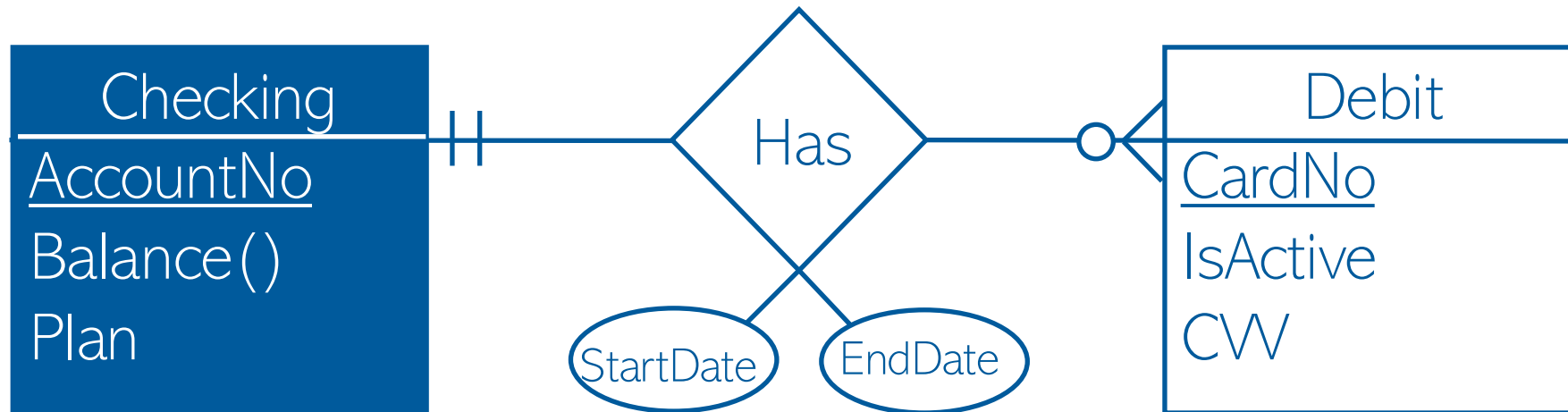
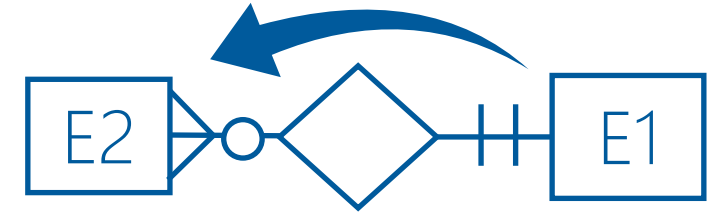
22

Relationship: Many-One (One-Many)



23

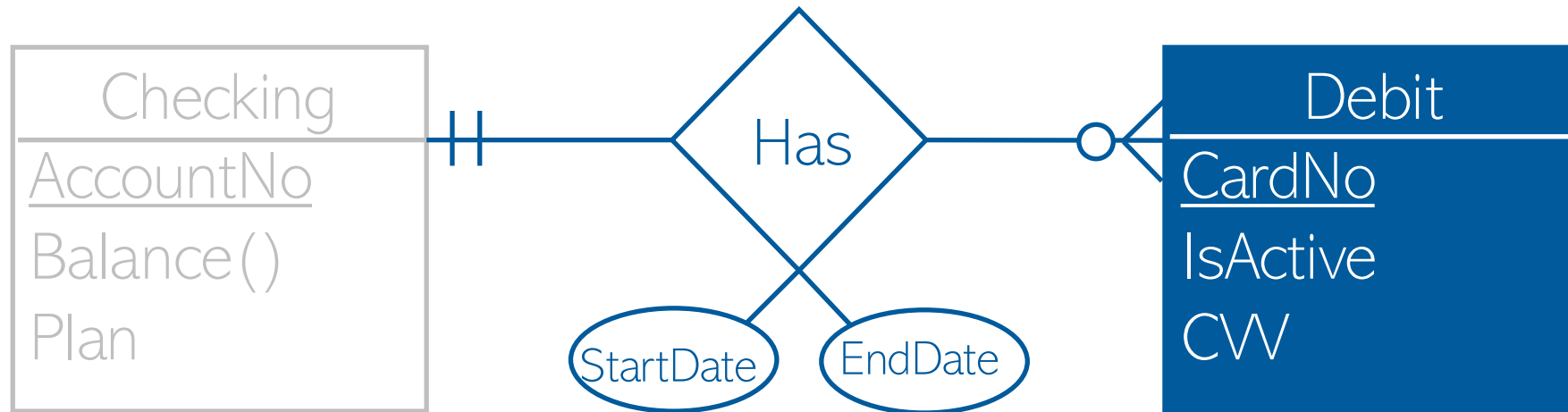
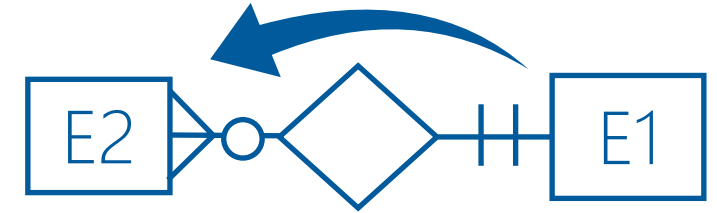
Relationship: Many-One (One-Many)



R1: Checking(AccountNo, Balance, Plan)

24

Relationship: Many-One (One-Many)

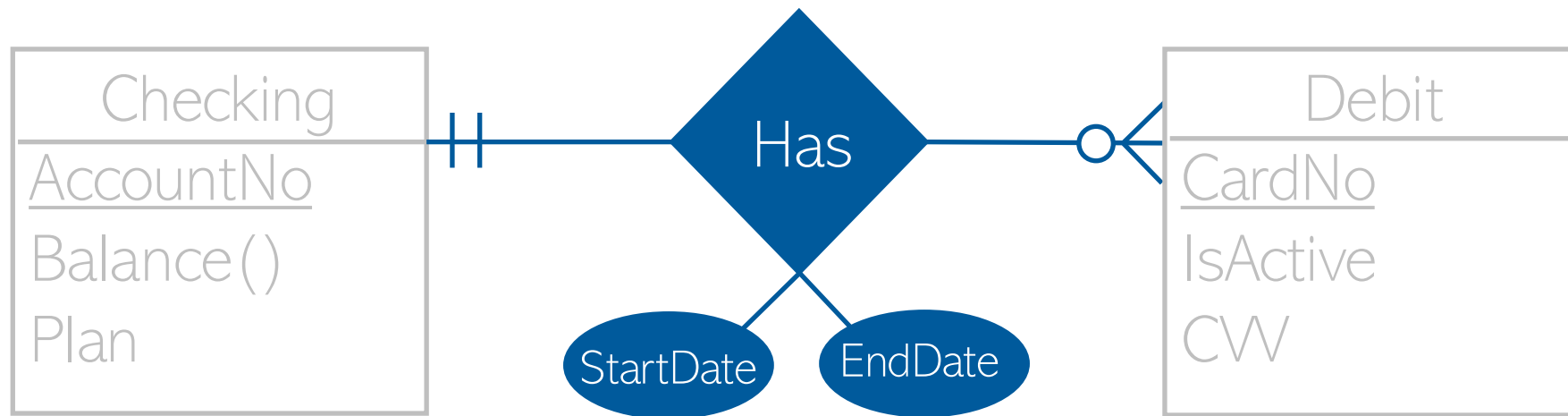
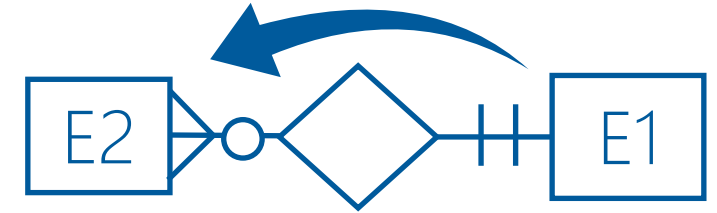


R1: Checking(AccountNo, Balance, Plan)

R2: Debit(CardNo, IsActive, CVV)

25

Relationship: Many-One (One-Many)

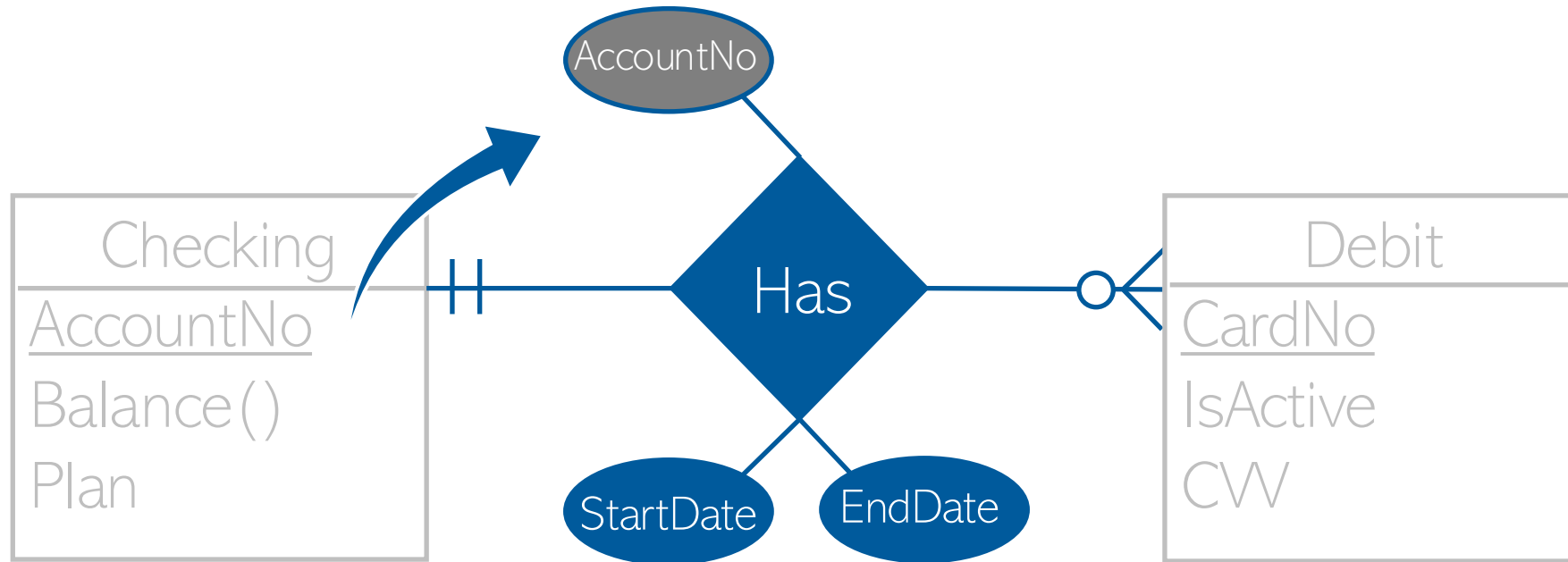
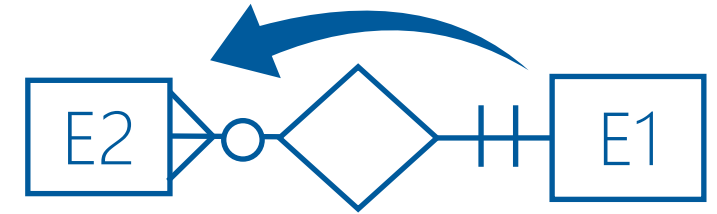


R1: Checking(AccountNo, Balance, Plan) ← ?

R2: Debit(CardNo, IsActive, CVV) ← ?

26

Relationship: Many-One (One-Many)



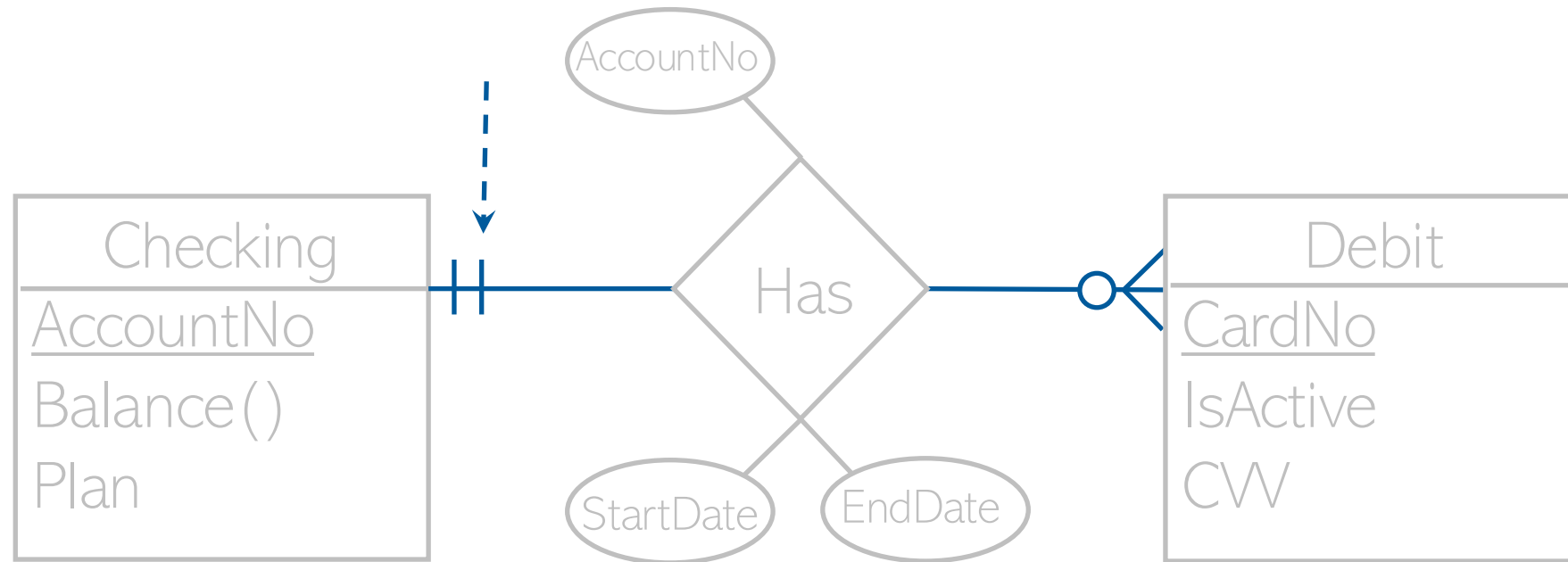
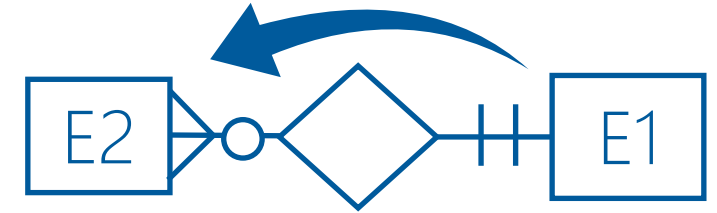
R1: Checking(AccountNo, Balance, Plan)

R2: Debit(CardNo, IsActive, CVV, Checking.AccountNo, StartDate, EndDate)

FK

27

Relationship: Many-One (One-Many)



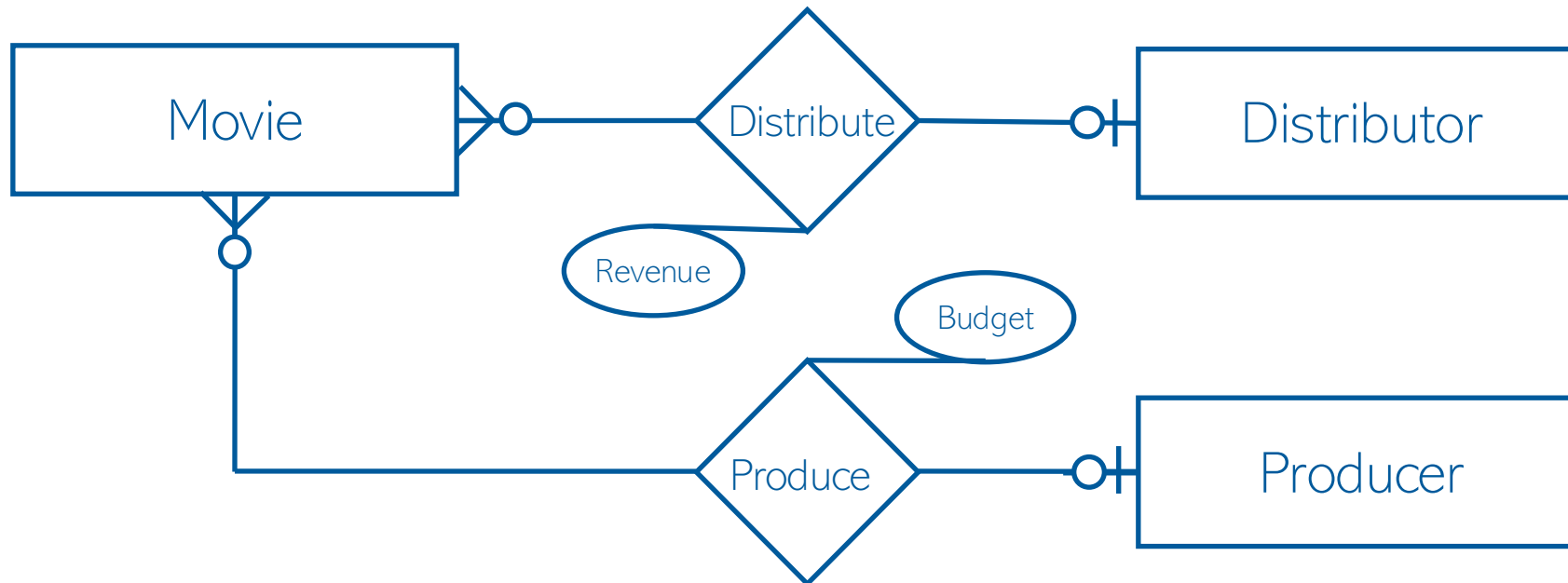
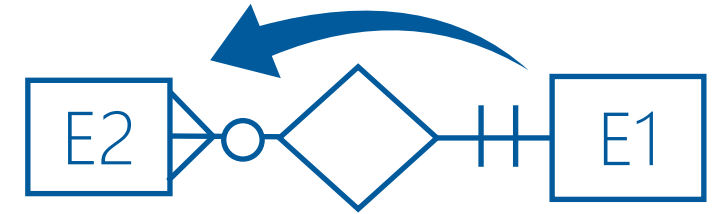
R1: Checking(AccountNo, Balance, Plan)

R2: Debit(CardNo, IsActive, CVV, Checking.AccountNo, StartDate, EndDate)

FK, Mandatory

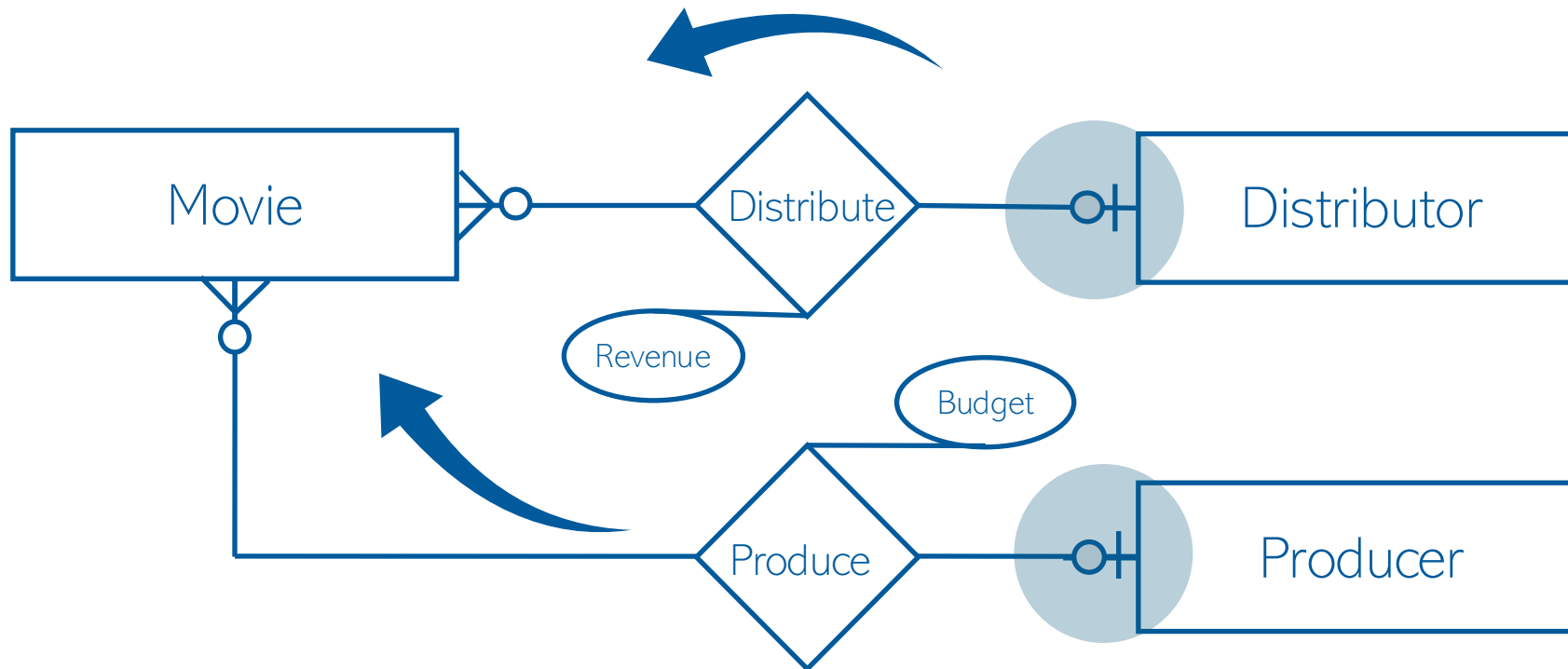
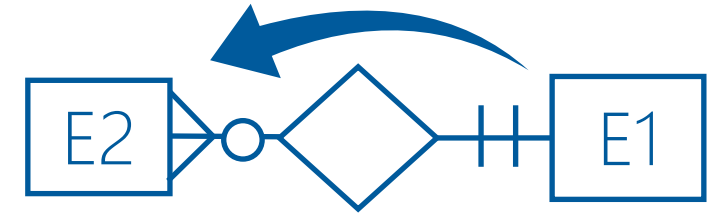
28

Relationship: Multiple Many-One



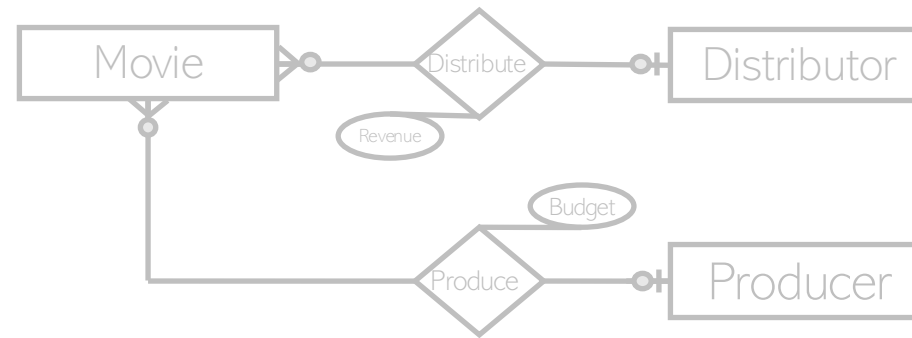
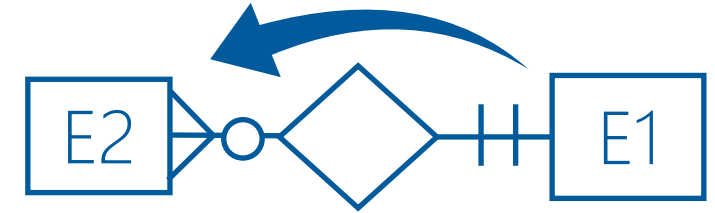
29

Relationship: Multiple Many-One



30

Relationship: Multiple Many-One



R1: Distributor(Name, Address, Phone, ...)

R2: Producer(Name, Studio, Address, Phone, ...)

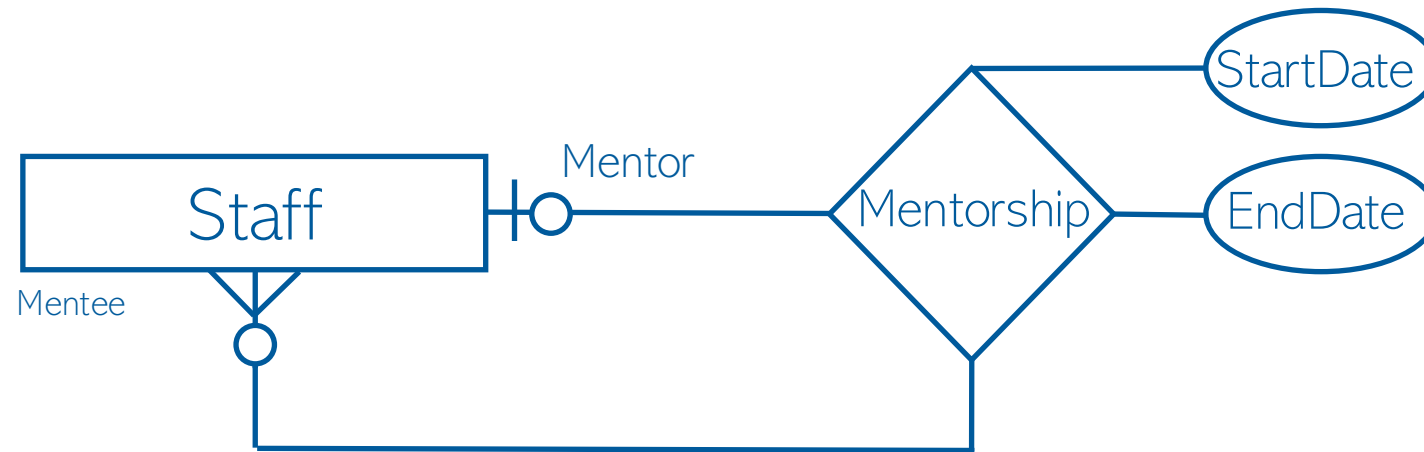
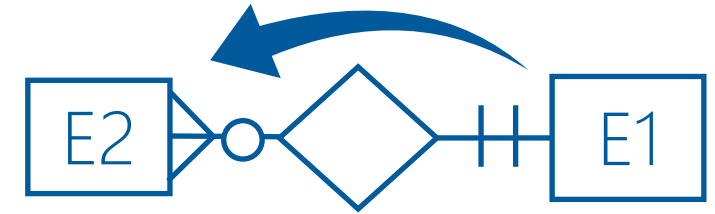
R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, **Distributor.Name**, Revenue, **Producer.Name**, Budget)

FK₁: {Distributor.Name} Optional (Nullable)

FK₂: {Producer.Name} Optional (Nullable)

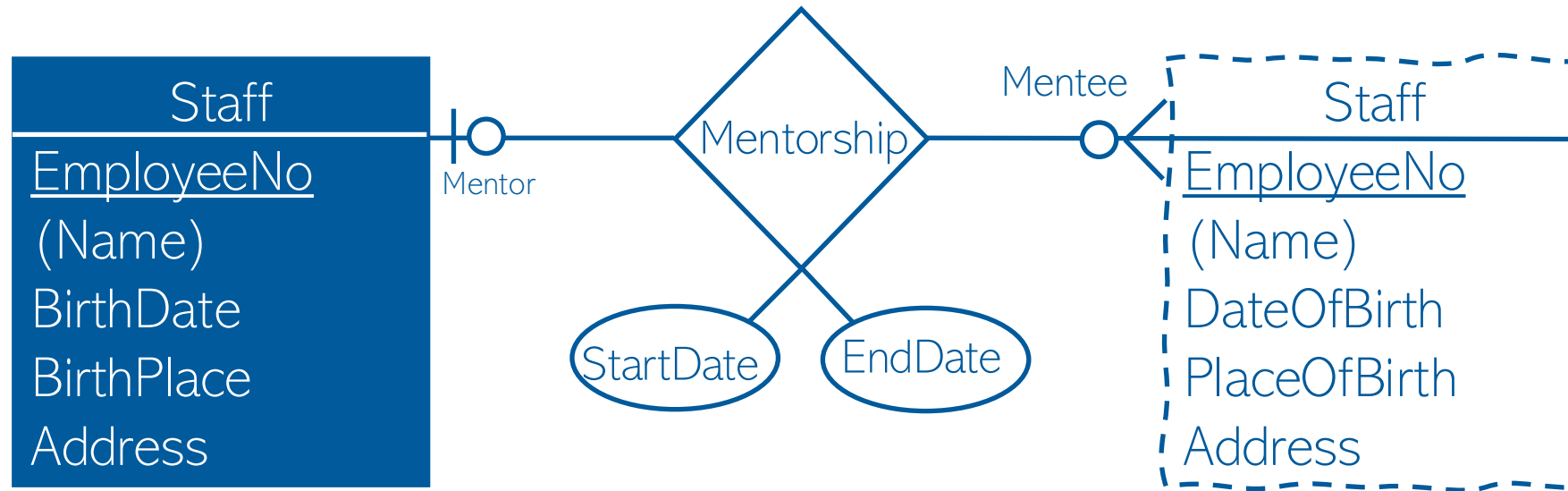
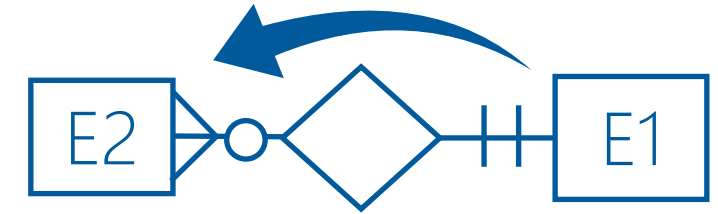
31

Relationship: Unary Many-One



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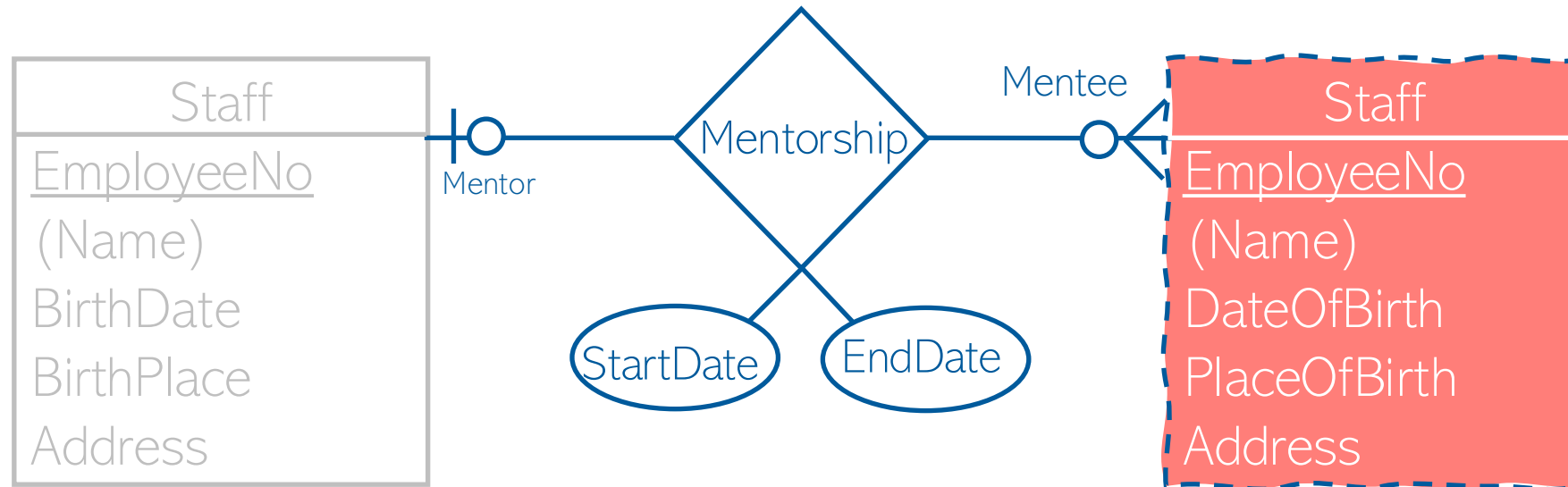
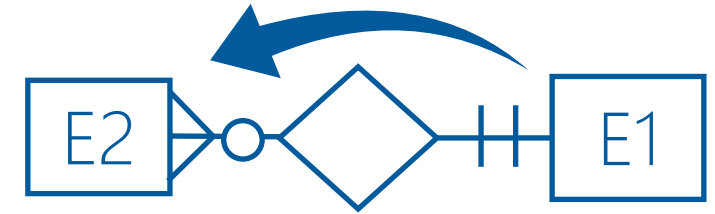
Relationship: Unary Many-One



R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

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Relationship: Unary Many-One

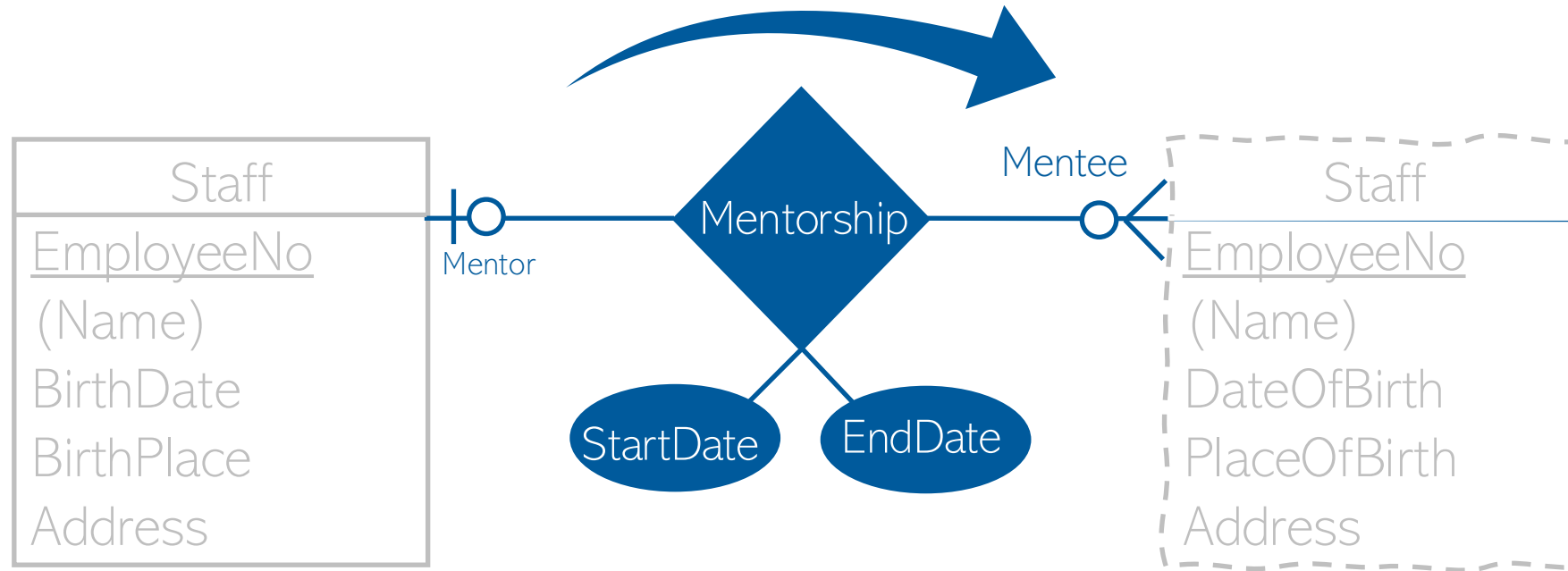
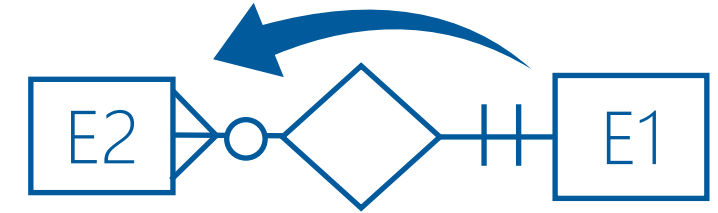


R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

34

Relationship: Unary Many-One



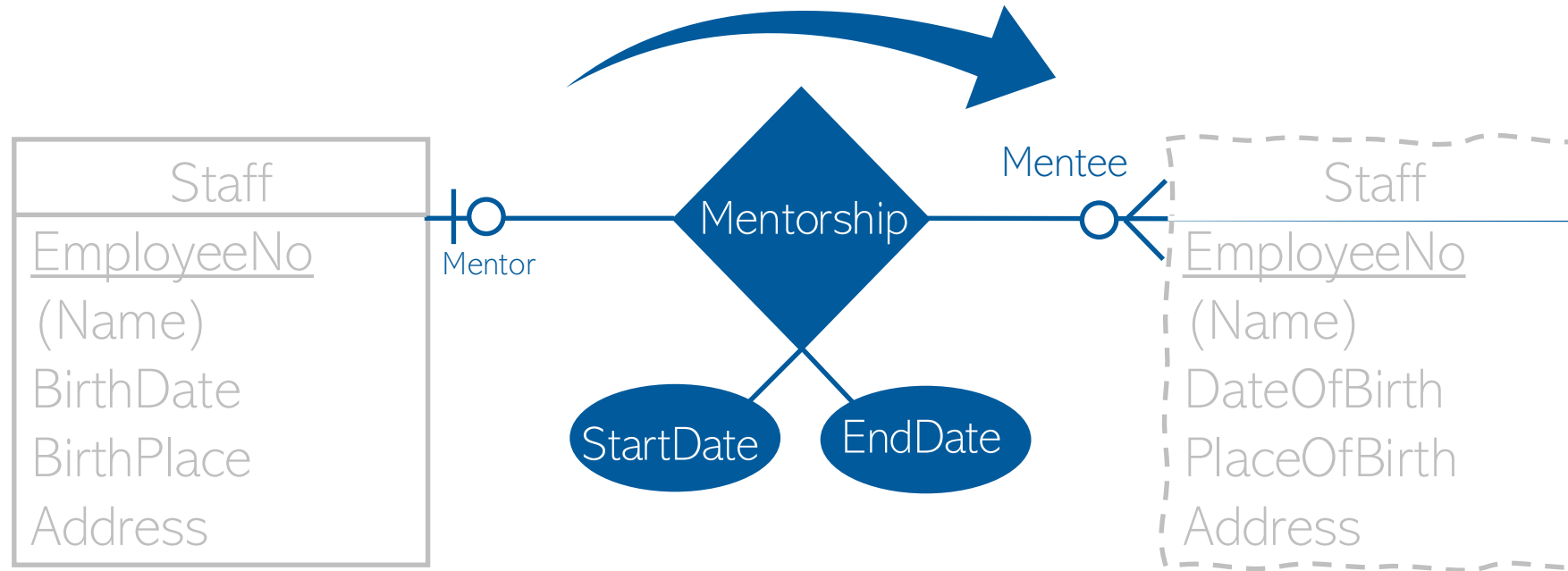
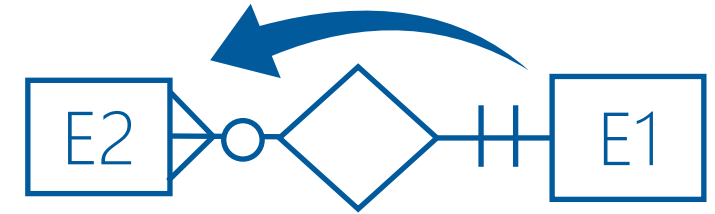
R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace,
Staff.EmployeeNo, StartDate, EndDate)

FK: Optional

35

Relationship: Unary Many-One



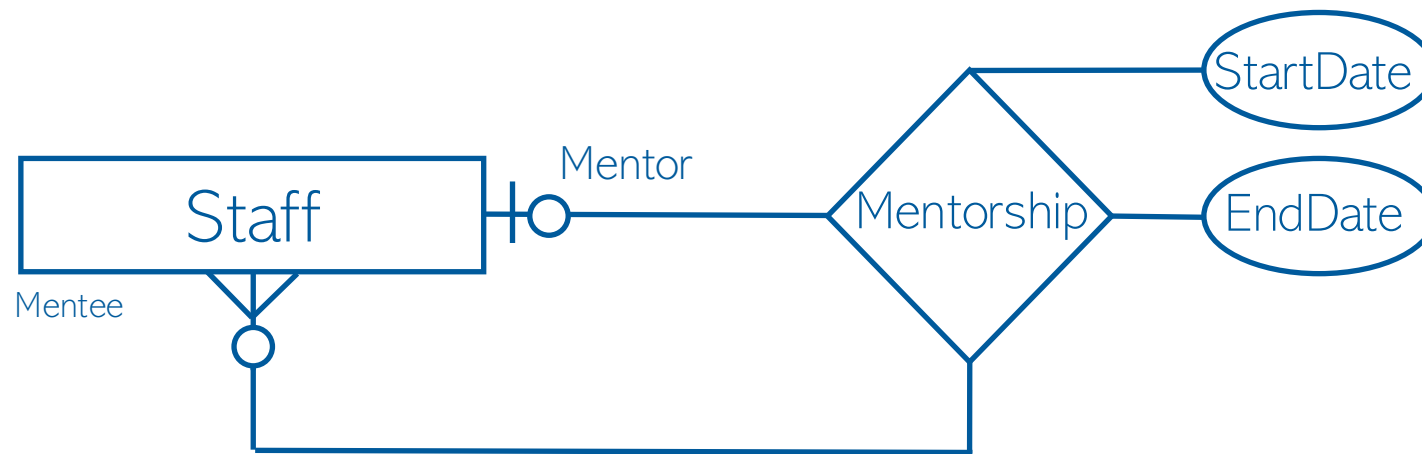
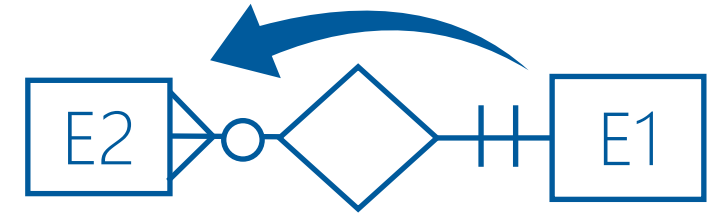
R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace,
MentorNo, StartDate, EndDate)

FK: Optional; To avoid possible name conflict, we use role name.

36

Relationship: Unary Many-One

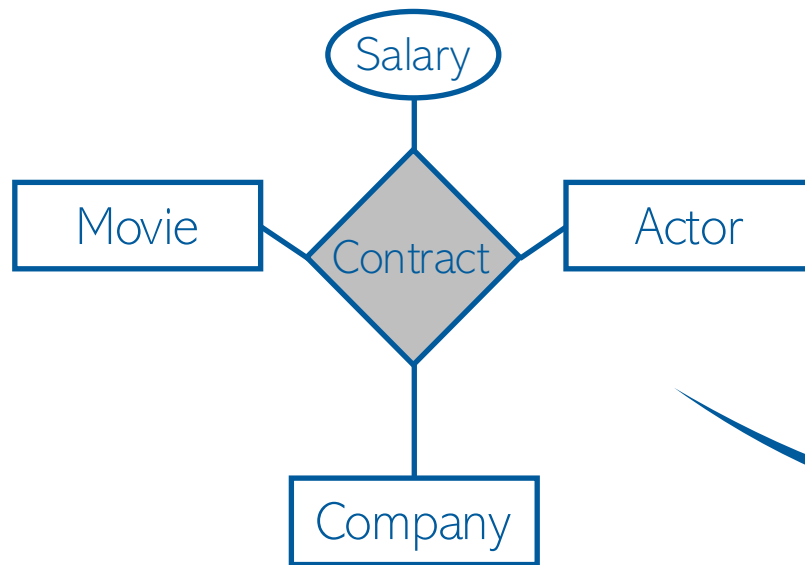
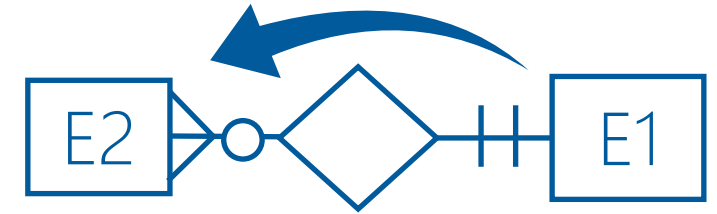


R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace, MentorNo, StartDate, EndDate)

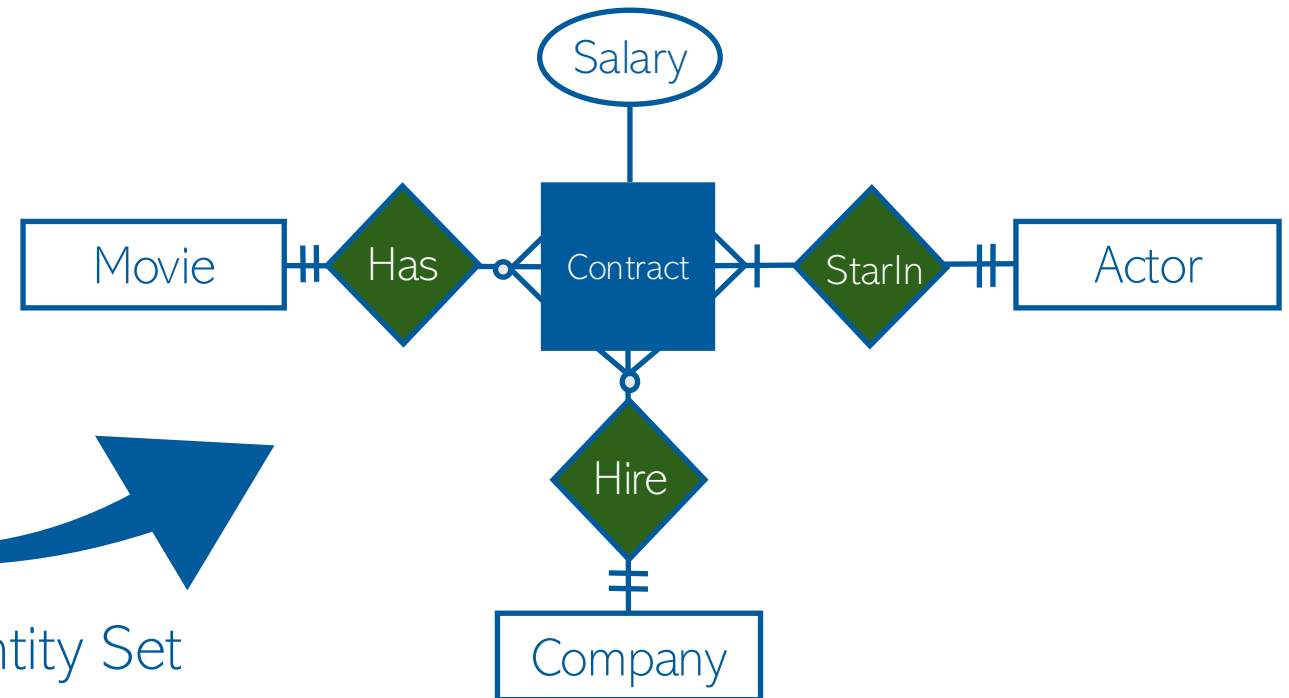
FK: {MentorNo} Optional

37

Relationship: Ternary Many-One

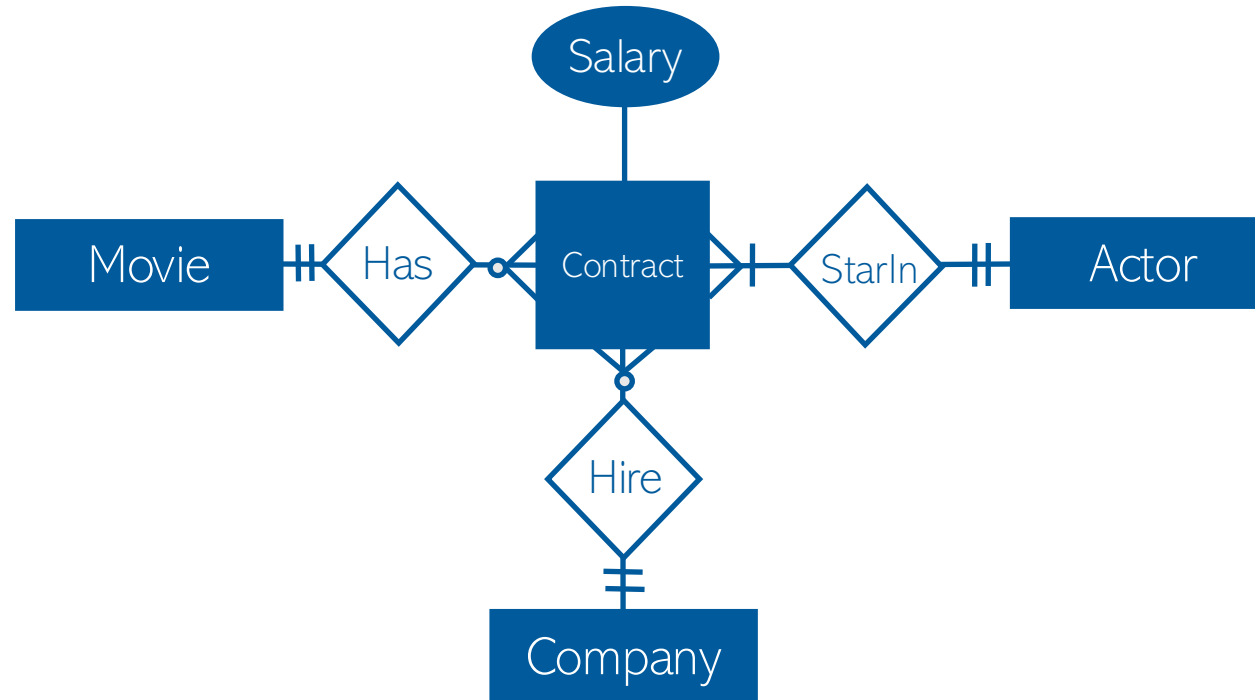
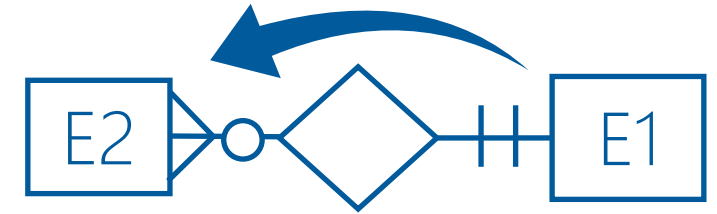


Connecting Entity Set



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Relationship: Ternary Many-One



R1: Company(Name, Address, Phone, ...)

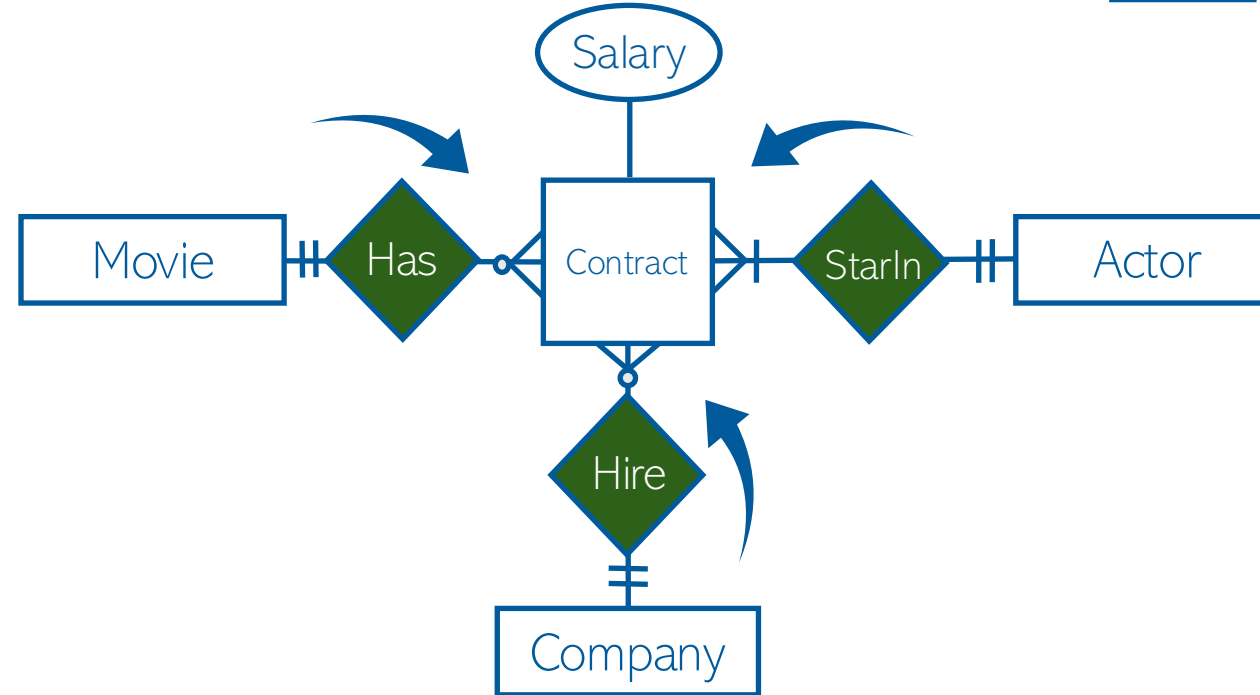
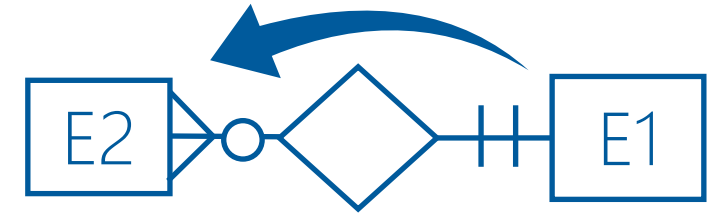
R2: Actor(FirstName, LastName, BirthDate, BirthPlace, ...)

R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime)

R4: Contract(Salary)

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Relationship: Ternary Many-One



R1: Company(Name, Address, Phone, ...)

R2: Actor(FirstName, LastName, BirthDate, BirthPlace, ...)

R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime)

R4: Contract(Salary

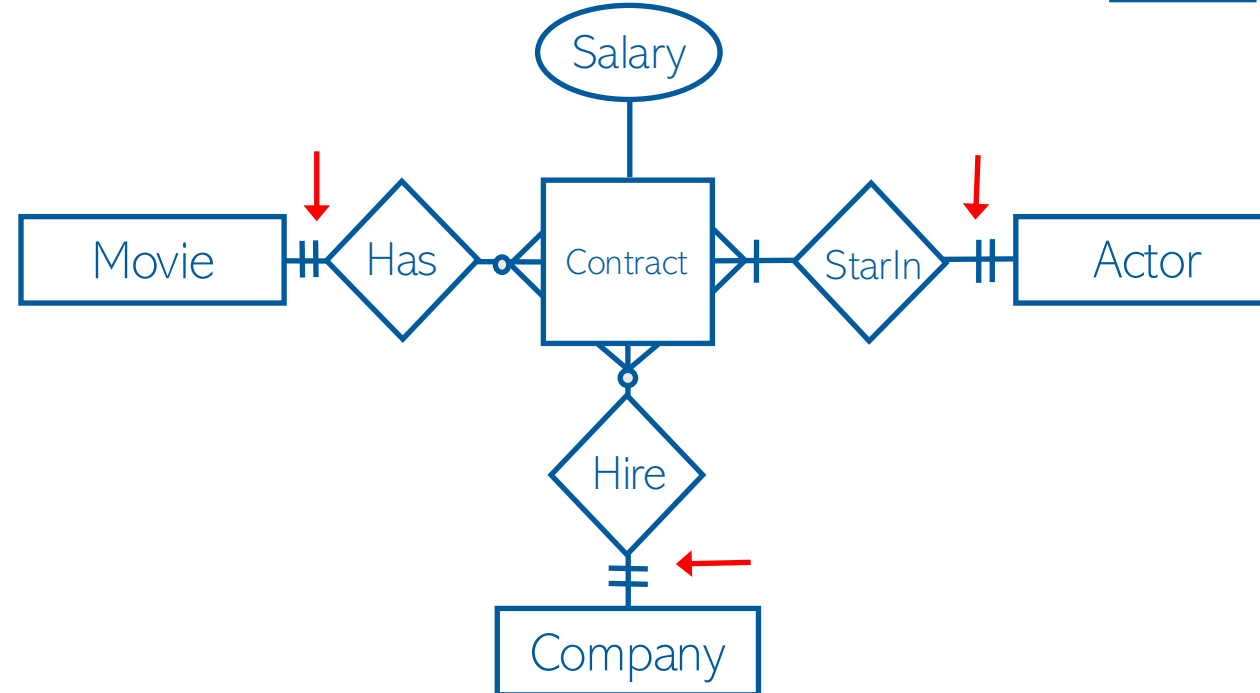
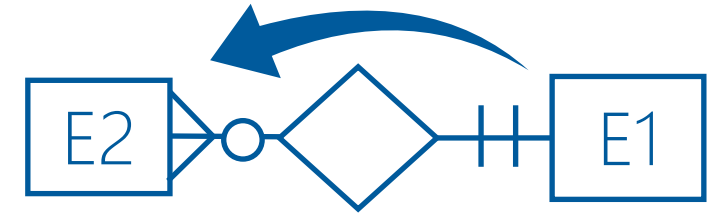
Company.Name, -----> FK₁

Actor.FirstName, Actor.LastName, Actor.BirthDate, Actor.BirthPlace -----> FK₂

Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay)-----> FK₃

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Relationship: Ternary Many-One



R1: Company(Name, Address, Phone, ...)

R2: Actor(FirstName, LastName, BirthDate, BirthPlace, ...)

R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime)

R4: Contract(Salary

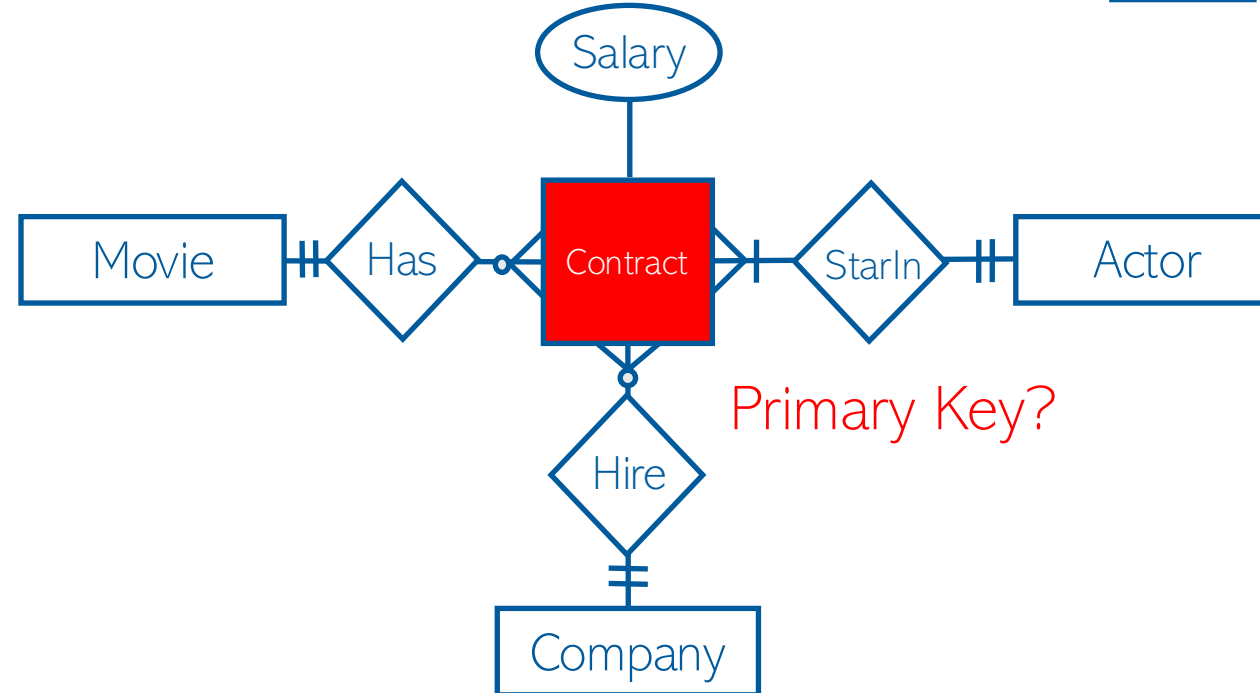
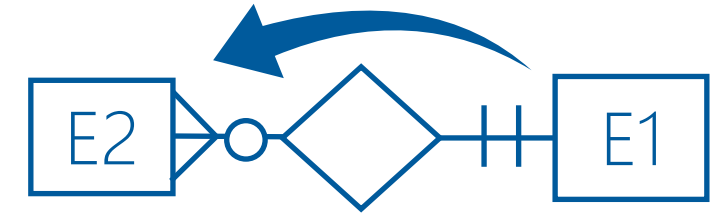
Company.Name, -----> FK₁, Mandatory

Actor.FirstName, Actor.LastName, Actor.BirthDate, Actor.BirthPlace -----> FK₂, Mandatory

Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay)-----> FK₃, Mandatory

41

Relationship: Ternary Many-One



R1: Company(Name, Address, Phone, ...)

R2: Actor(FirstName, LastName, BirthDate, BirthPlace, ...)

R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime)

R4: Contract(Salary

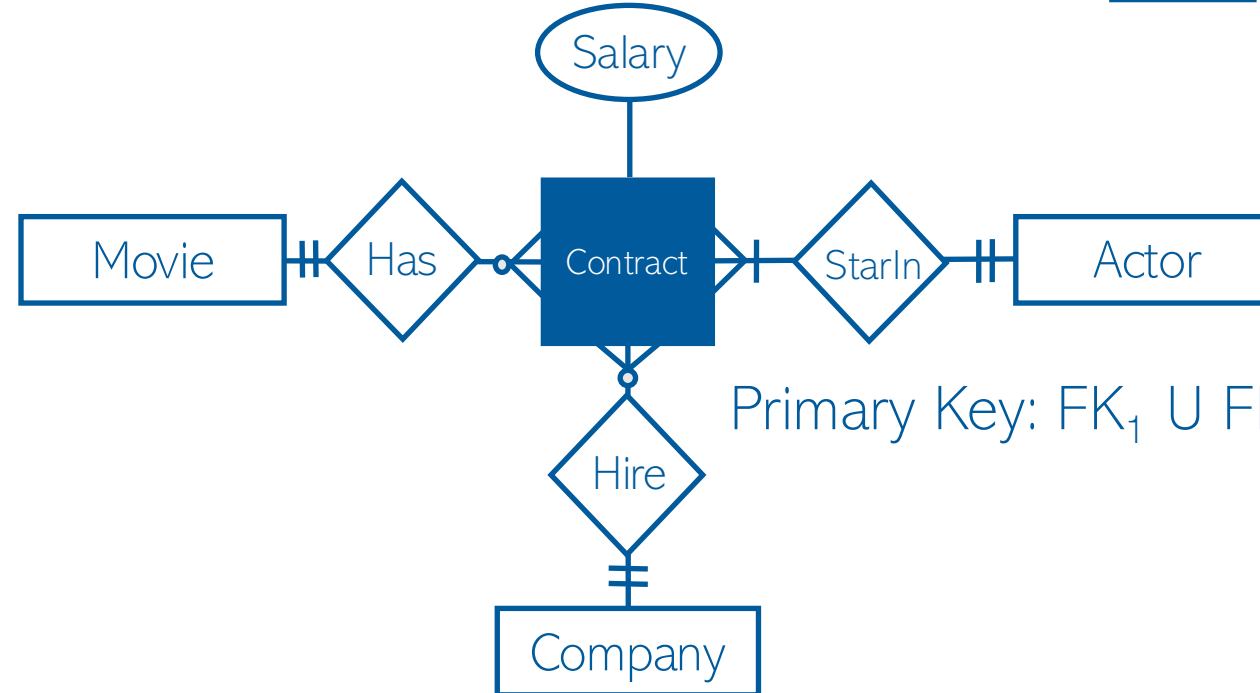
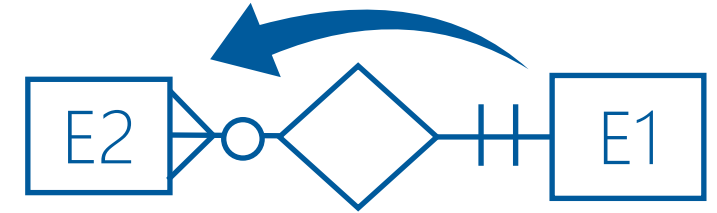
Company.Name, -----> FK₁, Mandatory

Actor.FirstName, Actor.LastName, Actor.BirthDate, Actor.BirthPlace -----> FK₂, Mandatory

Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay)-----> FK₃, Mandatory

42

Relationship: Ternary Many-One



Primary Key: $FK_1 \cup FK_2 \cup FK_3$

R1: Company(Name, Address, Phone, ...)

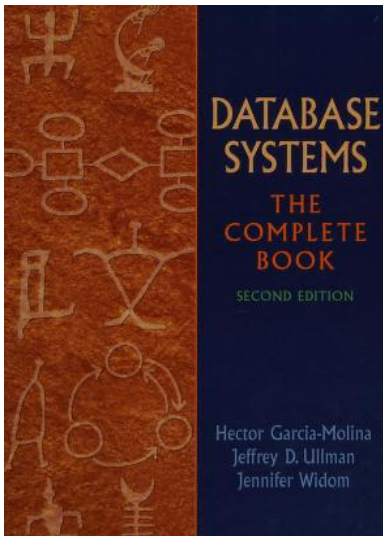
R2: Actor(FirstName, LastName, BirthDate, BirthPlace, ...)

R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime)

R4: Contract(Salary

Company.Name, -----> FK_1 , Mandatory
Actor.FirstName, Actor.LastName, Actor.BirthDate, Actor.BirthPlace -----> FK_2 , Mandatory
Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay)-----> FK_3 , Mandatory

Outline: W05A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

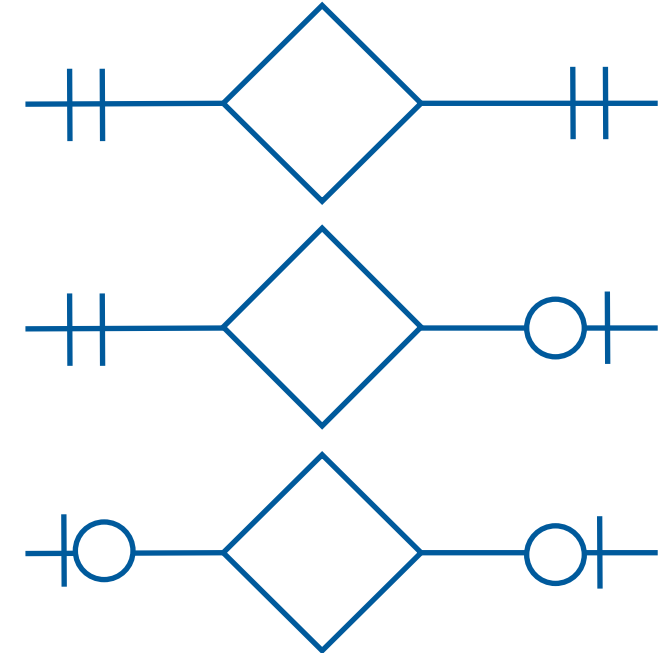
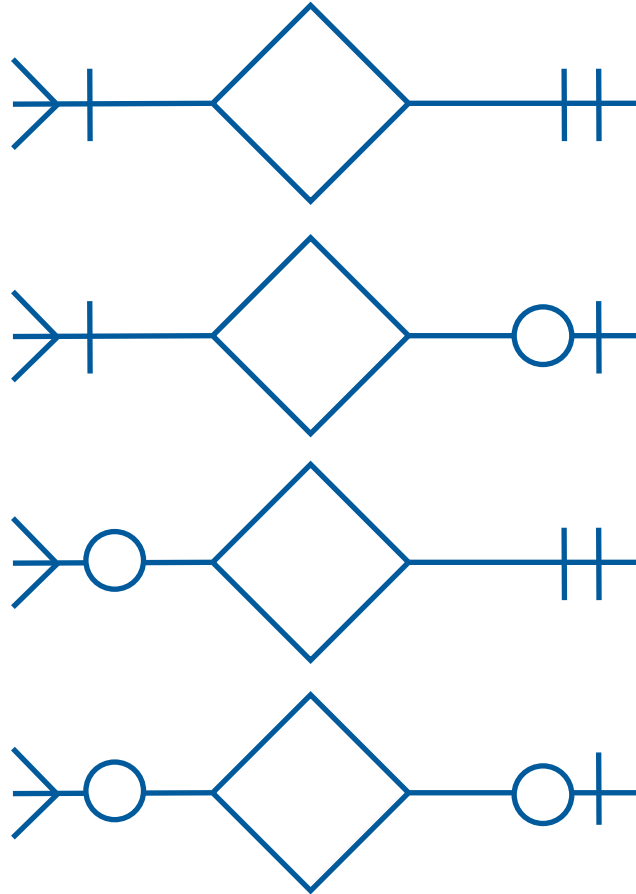
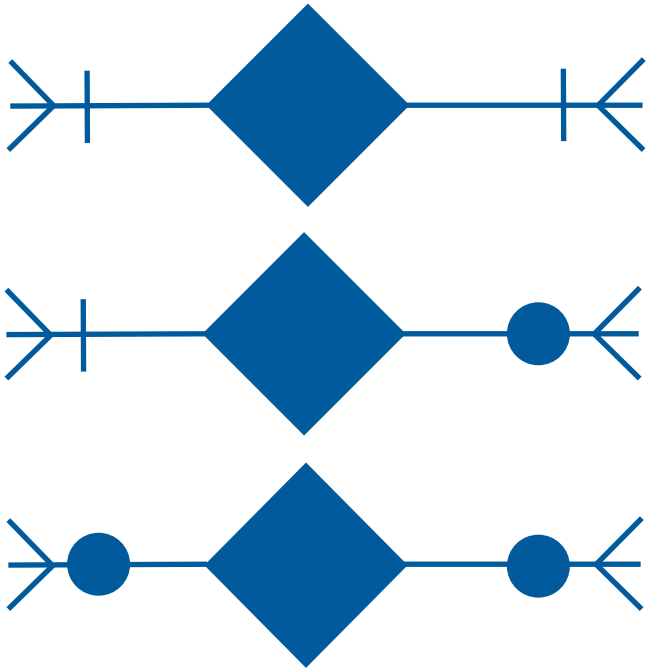
III) Relationship to Relation (Many-Many)

IIII) IS-A to Relation

Hands-On

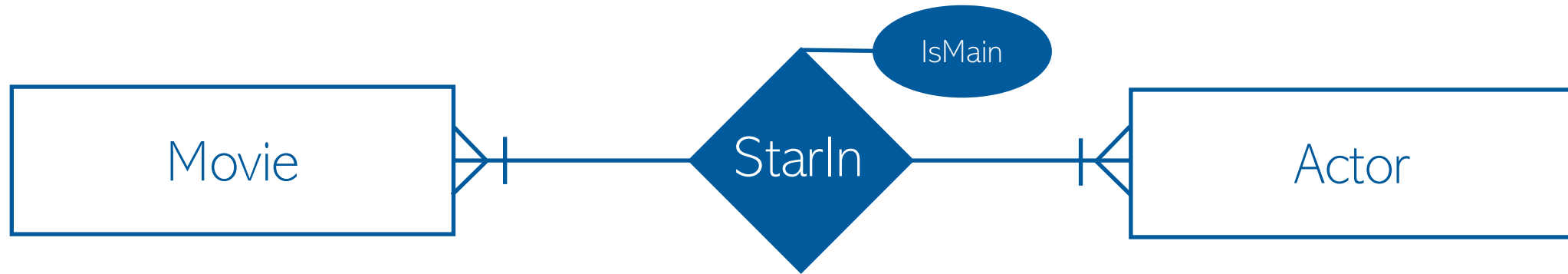
44

Outline: W05A



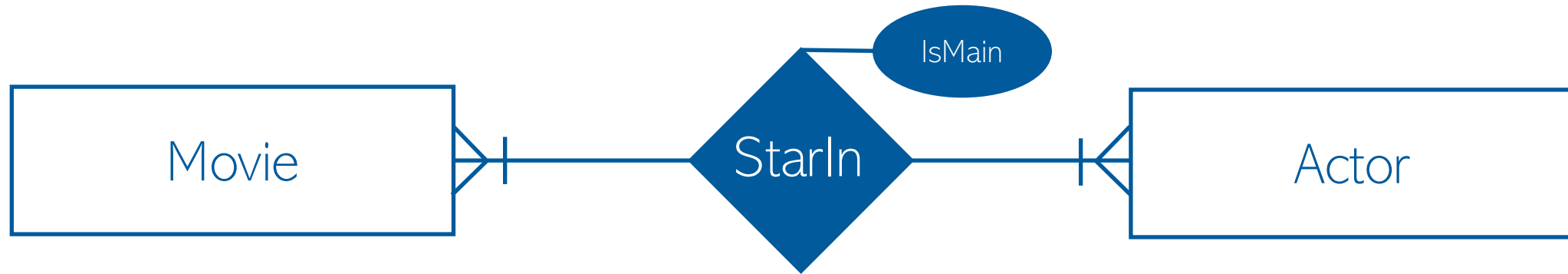
45

Relationship to Relations: Many-Many

 m_1 m_2 $m_1, a_1, 0$ $m_1, a_2, 1$ $m_1, a_3, 0$ $m_2, a_1, 1$ $m_2, a_3, 0$ a_1 a_2 a_3

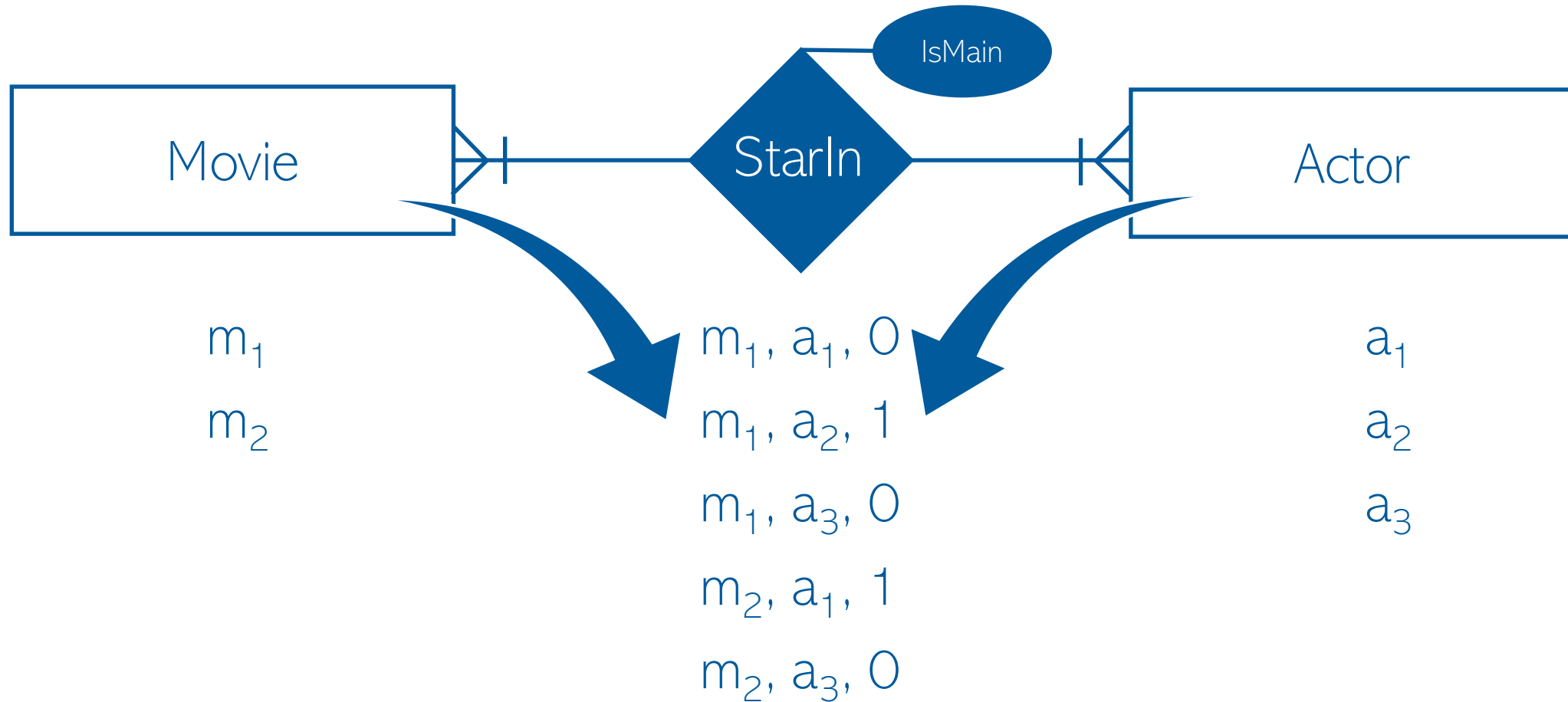
46

Relationship to Relations: Many-Many

 m_1, a_1, a_2, a_3 m_2, a_1, a_3 $m_1, a_1, 0$ $m_1, a_2, 1$ $m_1, a_3, 0$ $m_2, a_1, 1$ $m_2, a_3, 0$ a_1, m_1, m_2 a_2, m_1 a_3, m_1, m_2 

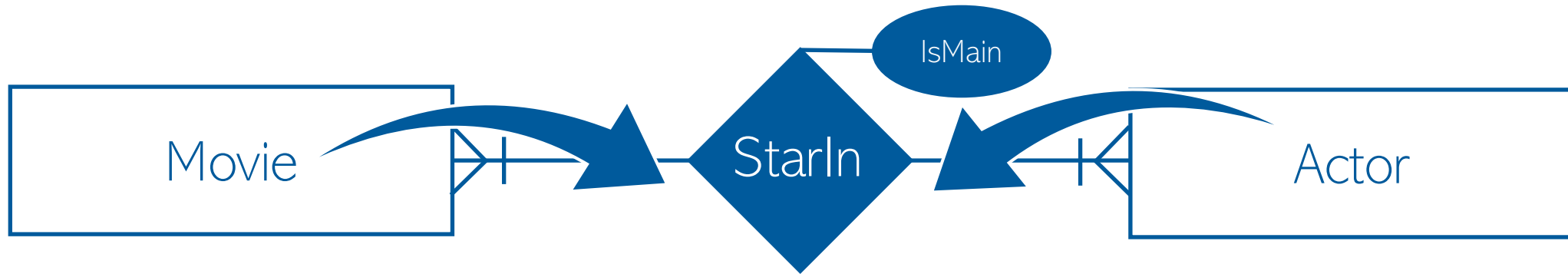
47

Relationship to Relations: Many-Many



48

Relationship to Relations: Many-Many



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

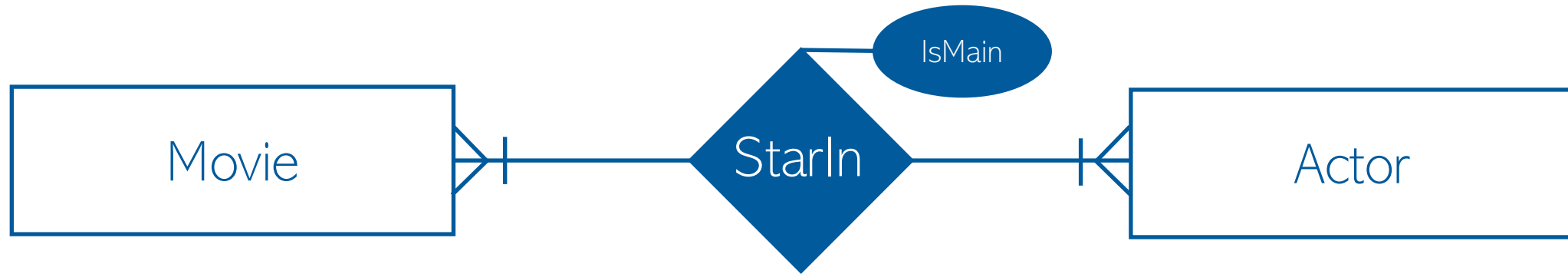
R2: Actor(FirstName, LastName, BirthDate, BirthPlace, Age)

R3: StarIn(IsMain,
 {Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay,
 Actor.FirstName, Actor.LastName, Actor.DateOfBirth},
)

FK₂ FK₁

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Relationship to Relations: Many-Many



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Actor(FirstName, LastName, BirthDate, BirthPlace, Age)

R3: StarIn(IsMain,
Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay,
Actor.FirstName, Actor.LastName, Actor.DateOfBirth)

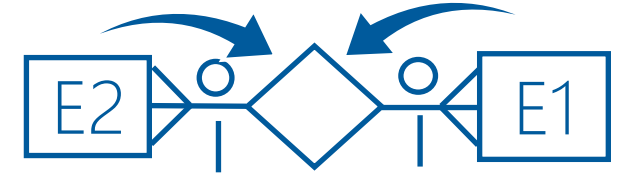
PK = FK₁ U FK₂

FK₂

FK₁

50

Relationship to Relations: Many-Many



Input: Many-Many relationship btw. E2 and E1, i.e.,

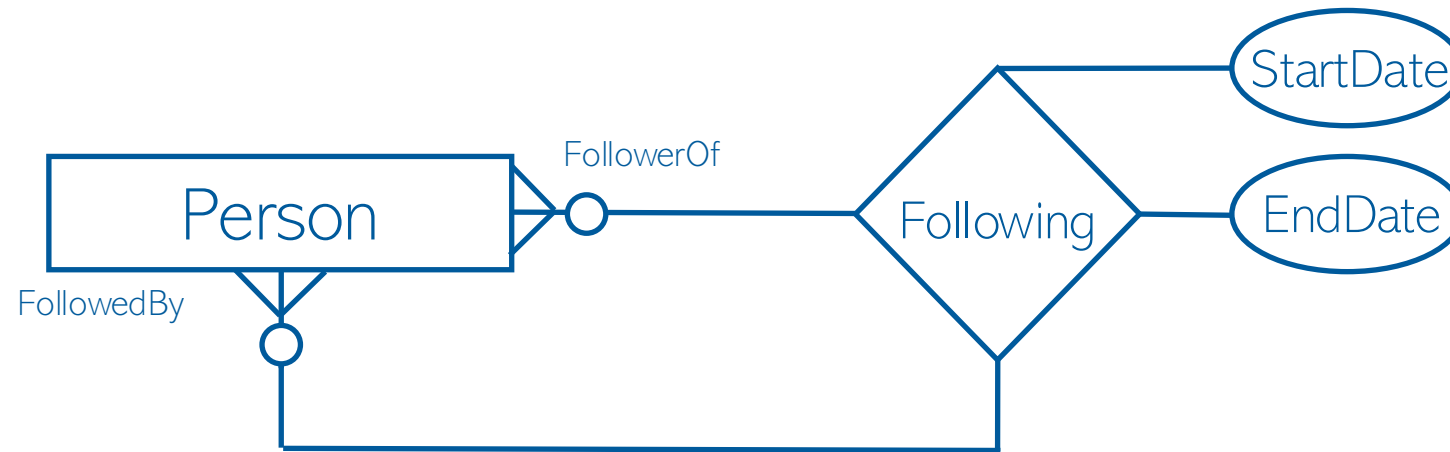
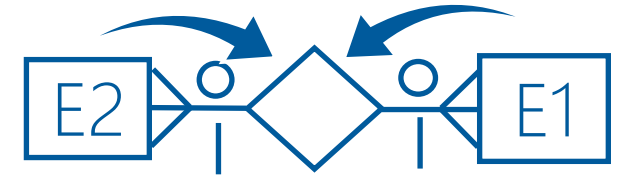
Output: Relations R1 for E1, R2 for E2, and R3 for relationship.

- 1) For E1, create relation R1 with the same attributes and keys as in E1
- 2) For E2, create relation R2 with the same attributes and keys as in E2
- 3) For many-many relationship set, **create new relation R3**
- 4) **[Foreign Key Set FK_1]** Add key set of E1 to R3
- 5) **[Foreign Key Set FK_2]** Add key set of E2 to R3
- 6) Add attributes of relationship set to R3
- 7) **[Primary Key Set]** Create key set for R3 from foreign keys (step 4 & 5) $PK = FK_1 \cup FK_2$

Herein, we do not care about ordinality! We fix it later.

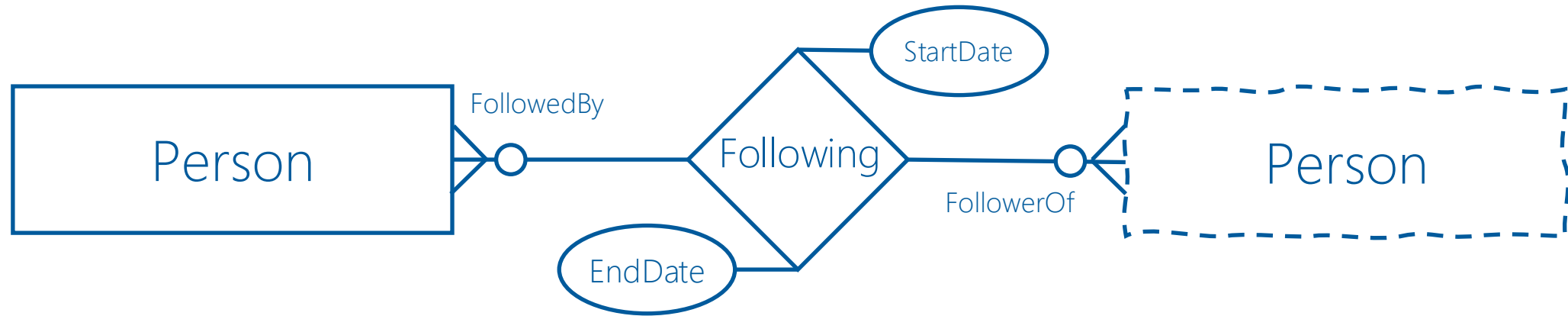
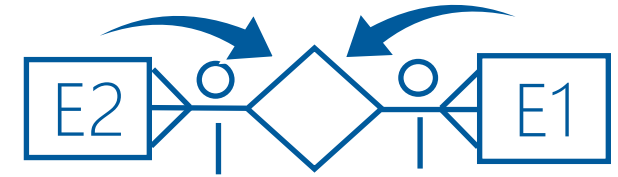
51

Relationship to Relations: Many-Many



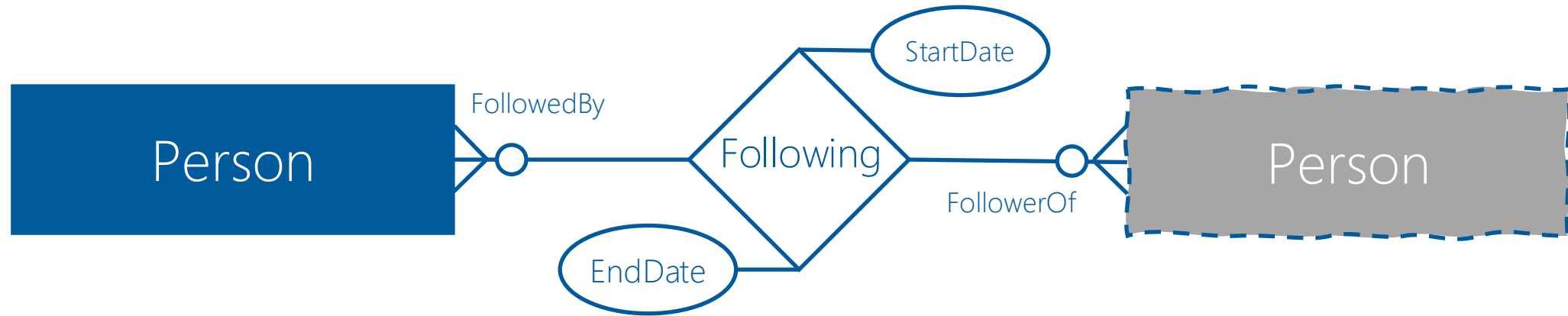
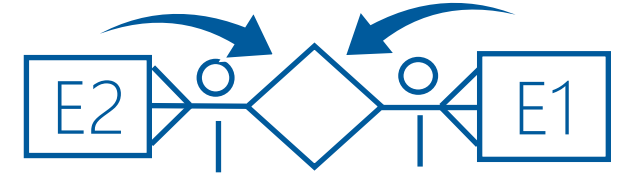
52

Relationship to Relations: Many-Many



53

Relationship to Relations: Many-Many

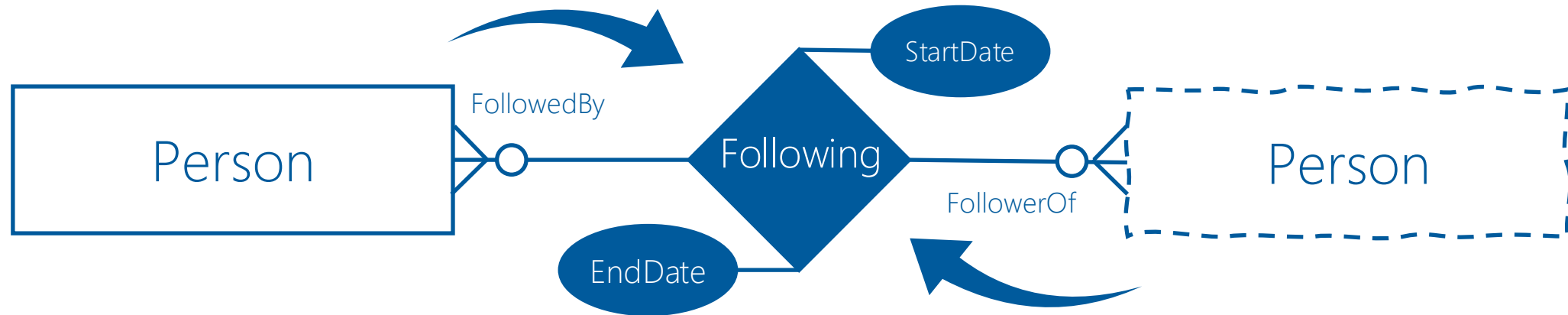
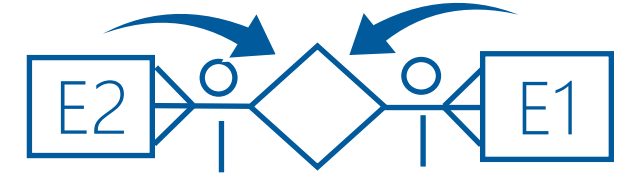


R1: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

R2: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

54

Relationship to Relations: Many-Many



R1: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

R2: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

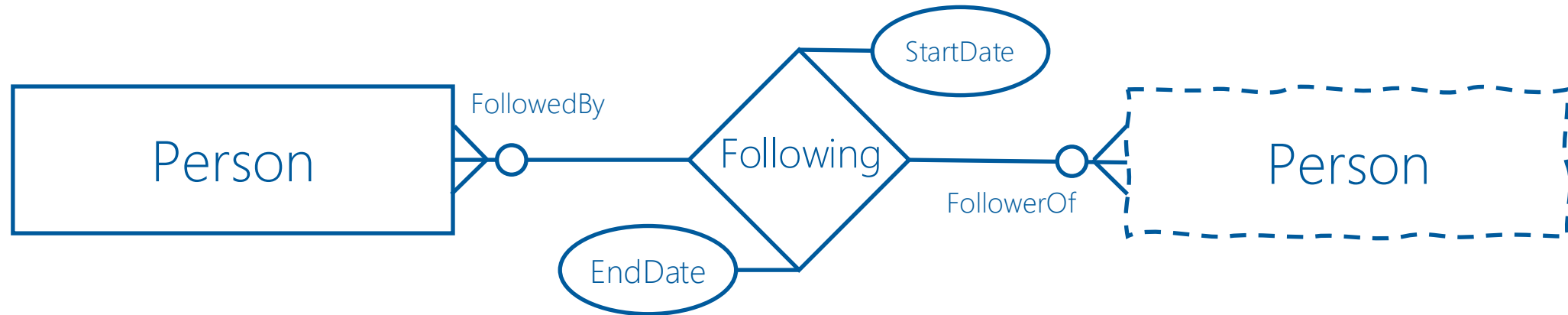
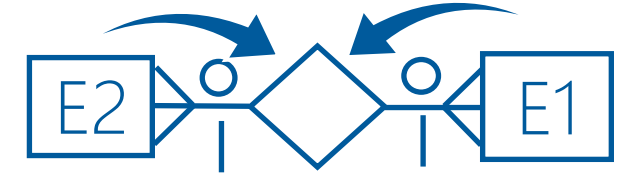
R3: Followership(FirstName, LastName, BirthDate, BirthPlace,
 FirstName, LastName, BirthDate, BirthPlace,
 StartDate, EndDate)

-----> FK₁

-----> FK₂

55

Relationship to Relations: Many-Many



R1: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

R2: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

R3: Followership(FirstName, LastName, BirthDate, BirthPlace,
 FirstName, LastName, BirthDate, BirthPlace,
 StartDate, EndDate)

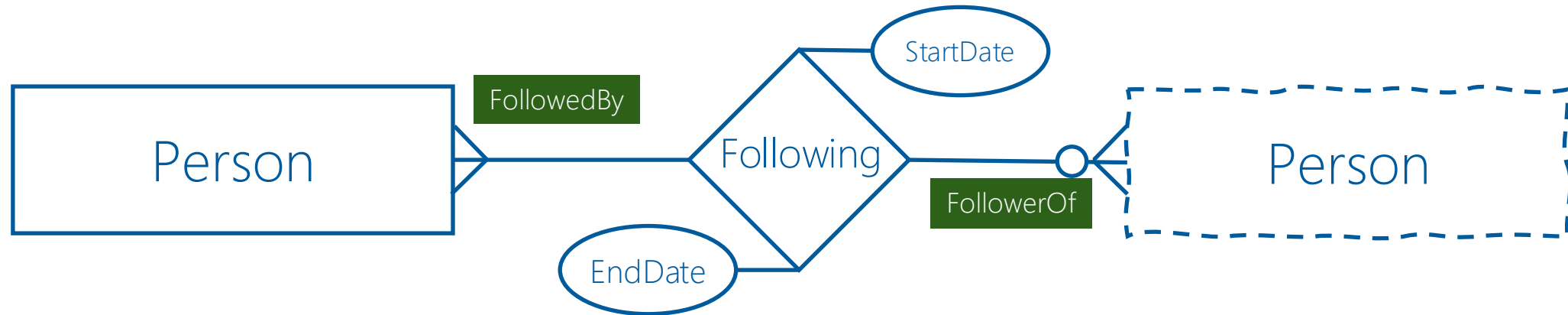
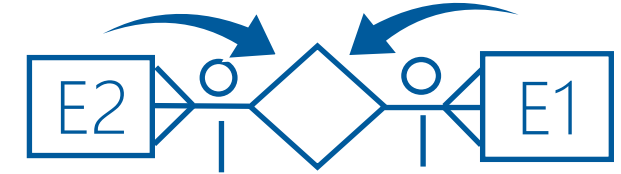
-----> FK₁

-----> FK₂

Name Conflict!

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Relationship to Relations: Many-Many



R1: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

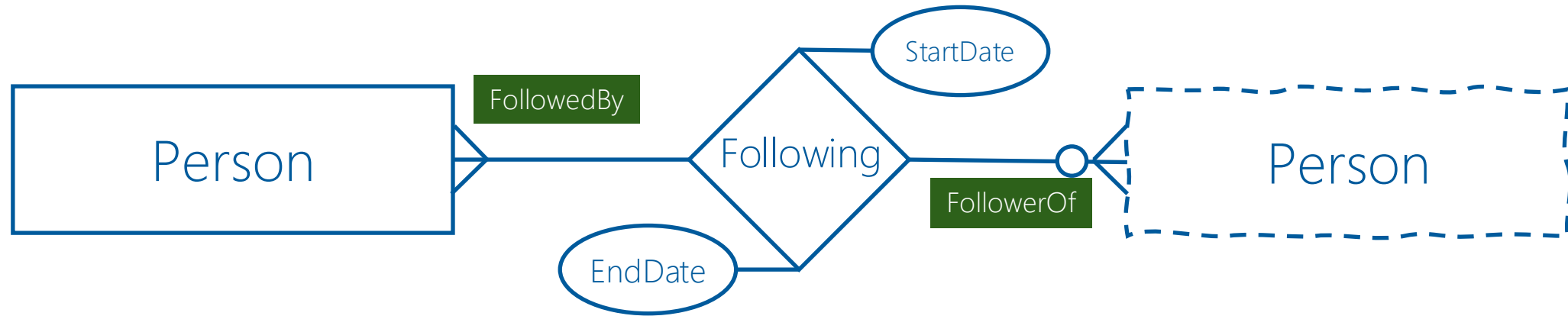
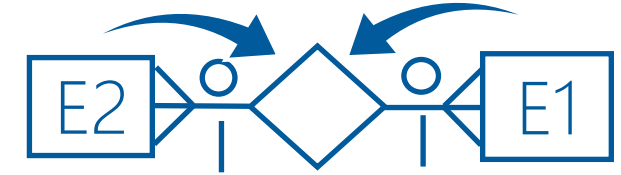
R2: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

R3: Followership(FollowerFirstName, FollowerLastName, FollowerBirthDate, FollowerBirthPlace, FK_1 ,
 FolloweeFirstName, FolloweeLastName, FolloweeBirthDate, FolloweeBirthPlace, FK_2 ,
 StartDate, EndDate)

Name Conflict Resolution: Role Playing

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Relationship to Relations: Many-Many



R1: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

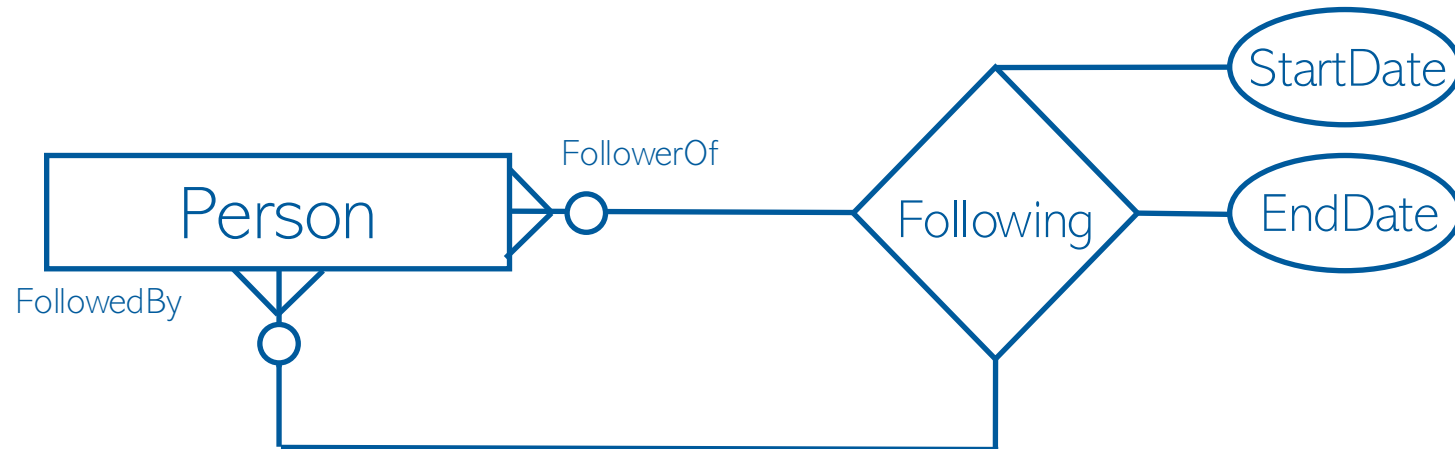
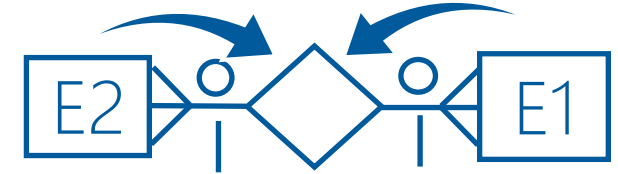
R2: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

R3: Followership(**FollowerFirstName**, **FollowerLastName**, **FollowerBirthDate**, **FollowerBirthPlace**, FK_1 ,
FolloweeFirstName, **FolloweeLastName**, **FolloweeBirthDate**, **FolloweeBirthPlace**, FK_2 ,
 StartDate, EndDate)

$PK = FK_1 \cup FK_2$ ("Should we include StartDate?")

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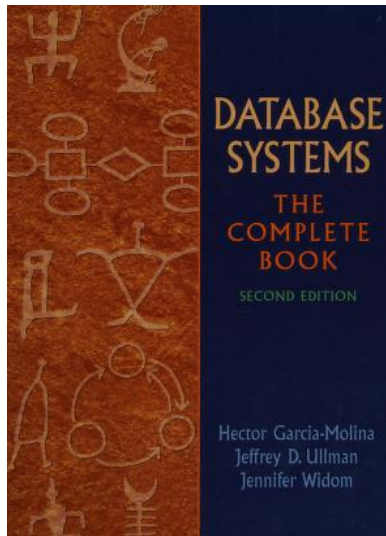
Relationship to Relations: Many-Many



R1: Person(FirstName, LastName, BirthDate, BirthPlace, Age, ...)

R3: Followership(FollowerFirstName, FollowerLastName, FollowerBirthDate, FollowerBirthPlace, FK₁,
FolloweeFirstName, FolloweeLastName, FolloweeBirthDate, FolloweeBirthPlace, FK₂,
 StartDate, EndDate)

Outline: W05A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

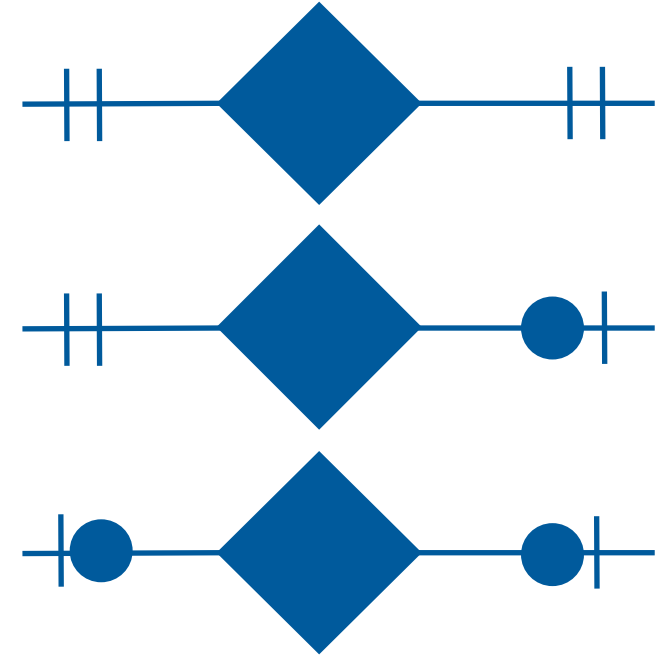
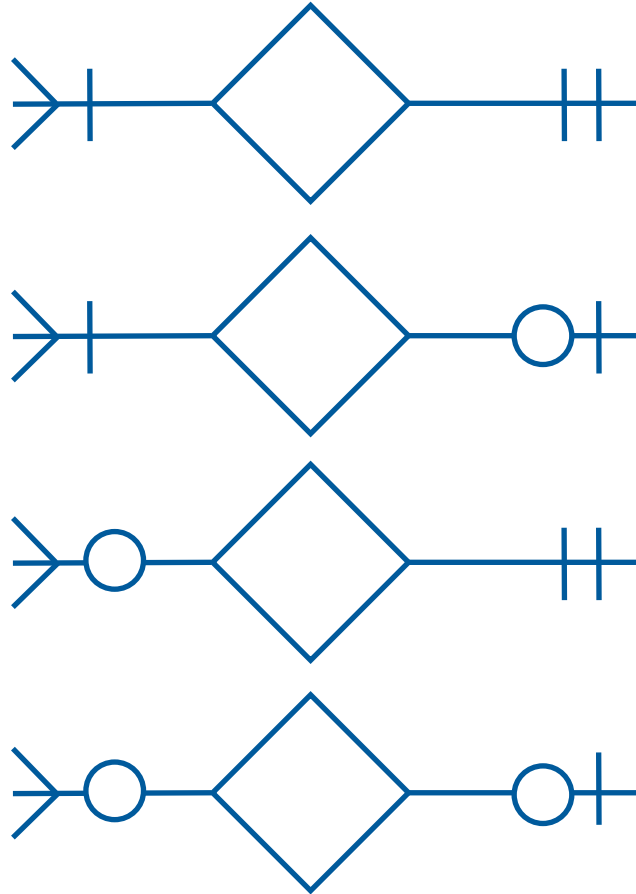
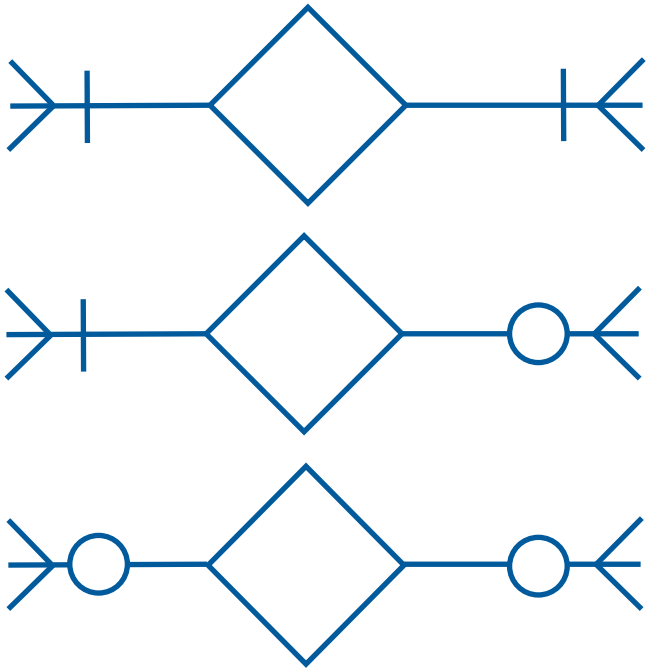
III) Relationship to Relation (One-One)

IIII) IS-A to Relation

Hands-On

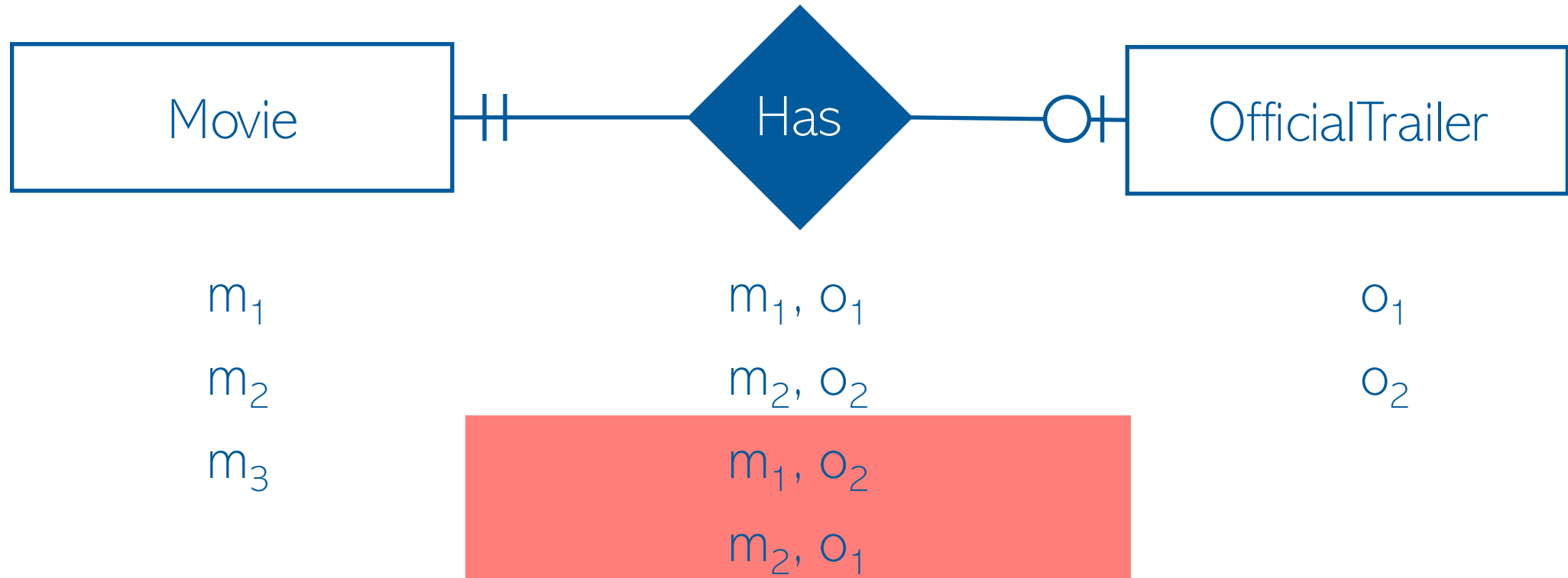
60

Outline: W05A



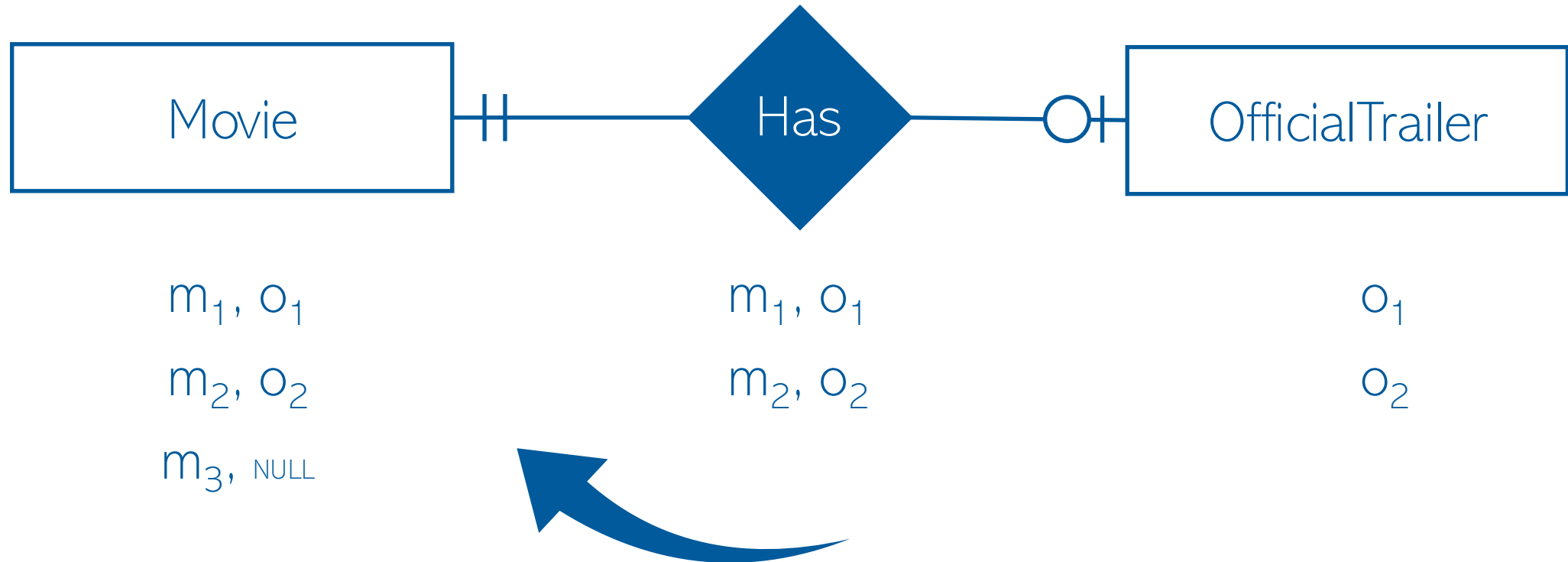
61

Relationship to Relations: One-One



62

Relationship to Relations: One-One (Approach A)



63

Relationship to Relations: One-One (Approach A)



R1: OfficialTrailer(Url, Content, RunningTime)

R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, OfficialTrailer.Url)

FK: Optional

64

Relationship to Relations: One-One (Approach A)



R1: OfficialTrailer(Url, Content, RunningTime)

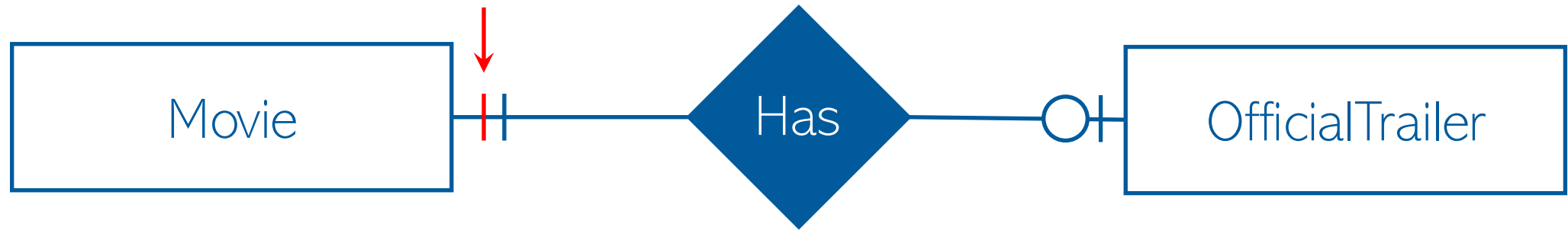
R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, OfficialTrailer.Url)

FK: Optional

There is a problem! Where? Solution?

65

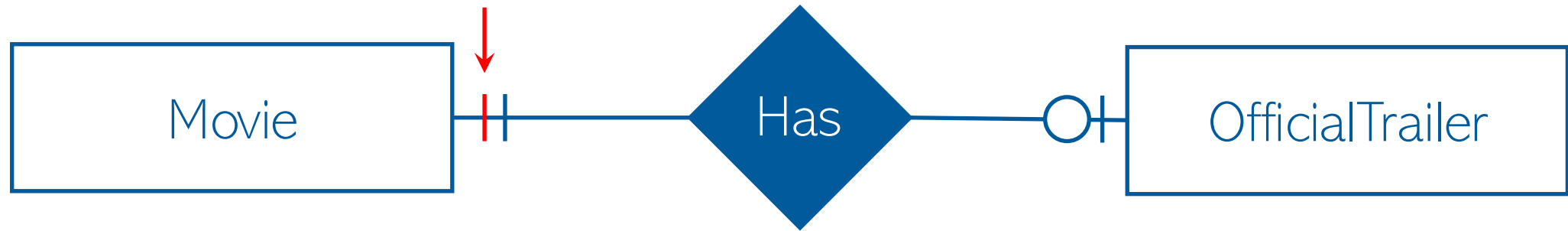
Relationship to Relations: One-One (Approach A)

 m_1, o_1 m_2, o_2 m_3, NULL m_2, o_1 m_1, o_1 m_2, o_2 o_1 o_2

An official trailer belongs to one and only one movie, but nothing prevents this error!

66

Relationship to Relations: One-One (Approach A)



m_1	o_1
m_2	o_2
m_3	NULL
m_2	o_1

 m_1, o_1 o_1 m_2, o_2 o_2

An official trailer belongs to one and only one movie. If we make this domain (column) unique ...

67

Relationship to Relations: One-One (Approach A)



R1: OfficialTrailer(Url, Content, RunningTime)

R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, OfficialTrailer.Url)

PK={Title, ReleaseYear, ReleaseMonth, ReleaseDay} U OfficialTrailer.Url

FK: Optional

By being part of PK?

Does not solve the issue. Foremost, OfficialTrailer.Url is optional!

Even if it was mandatory, an official trailer can still belong to more than one movie ...

68

Relationship to Relations: One-One (Approach A)



R1: OfficialTrailer(Url, Content, RunningTime)

R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, OfficialTrailer.Url)

SK=Secondary Key={OfficialTrailer.Url}

FK: Optional

Being an alternative secondary PK? Yes, but optional.



Divine Intervention ..., Pulp Fiction, Quintine Tarantino, 1994

<https://www.youtube.com/watch?v=l2oaQ3Qv7pQ>

70

Relationship to Relations: For Any Relations

Candidate Key (CK):

A key set whose attributes can accept no value (NULL).

In Primary Key, all its attributes must have value, but CK's attributes could be optional.

So, relational model allows multiple key sets for a relation (table). It is *not* called Secondary Key set ... It's called Candidate Key (CK).

ER does not have it.

71

Relationship to Relations: One-One (Approach A)



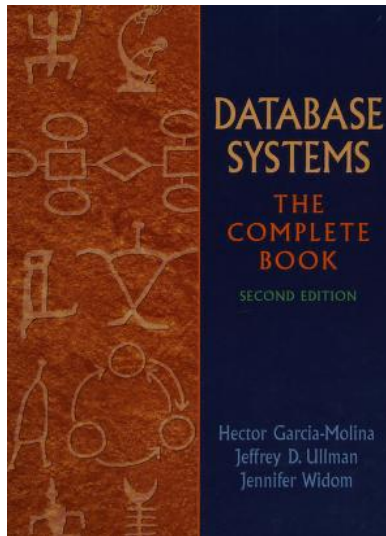
R1: OfficialTrailer(Url, Content, RunningTime)

R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, OfficialTrailer.Url)

CK=FK={OfficialTrailer.Url}

FK: Optional

Outline: W05A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

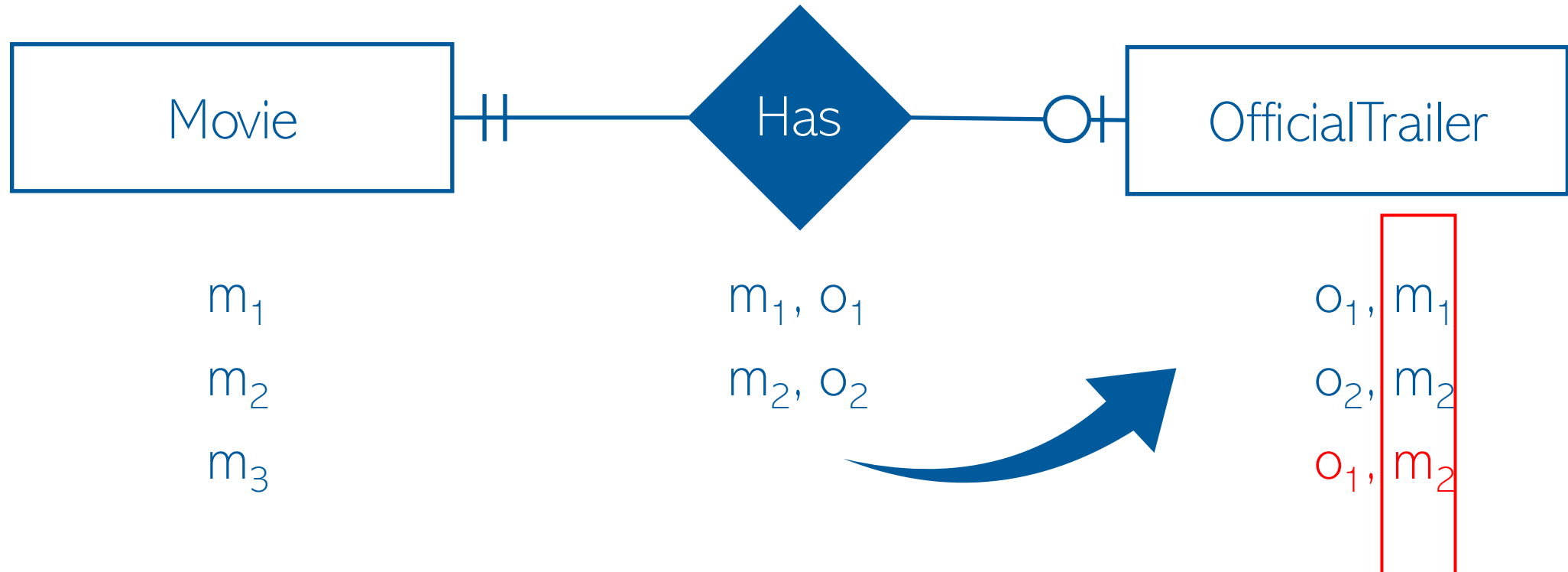
III) Relationship to Relation (One-One)

IIII) IS-A to Relation

Hands-On

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Relationship to Relations: One-One (Approach B)



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Relationship to Relations: One-One (Approach B)

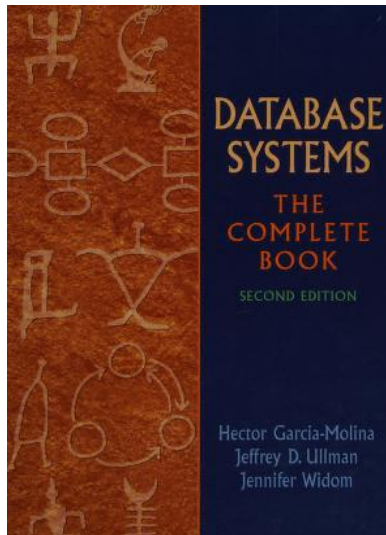


FK: Mandatory

R1: OfficialTrailer(Url, Content, ..., Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay)
CK=FK= {Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay}

R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

Outline: W05A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

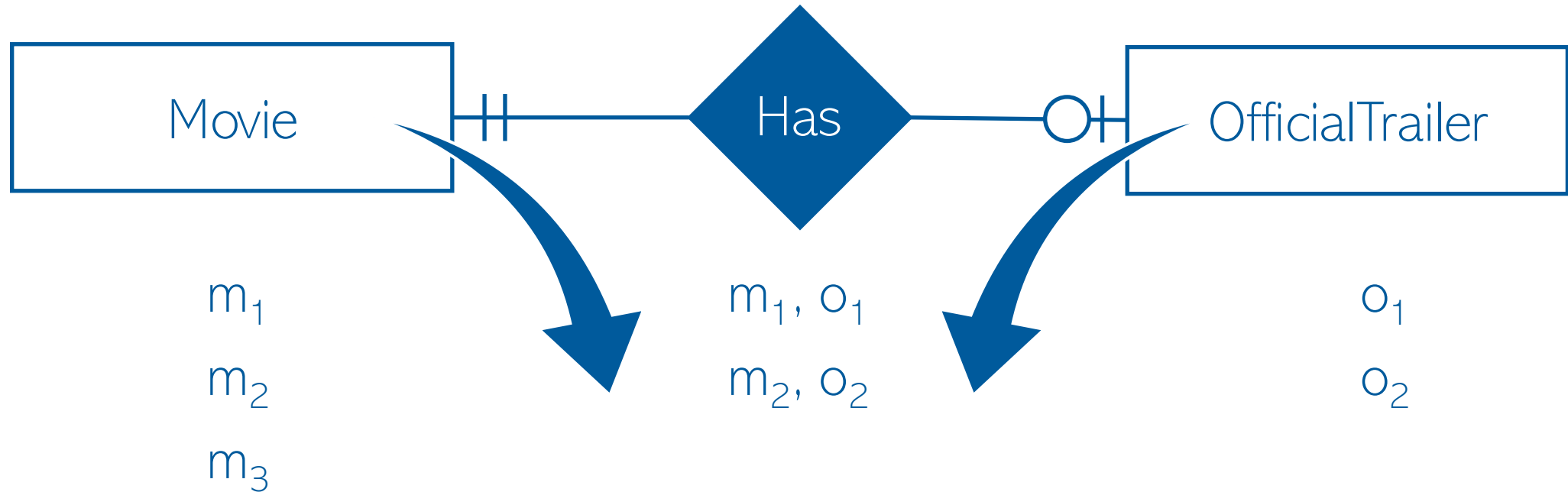
III) Relationship to Relation (One-One)

IIII) IS-A to Relation

Hands-On

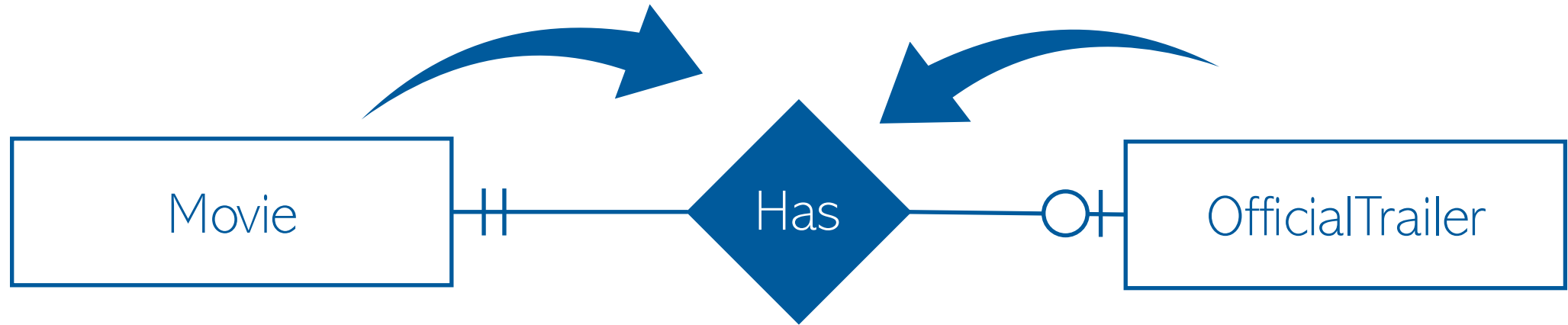
76

Relationship to Relations: One-One (Approach C)



77

Relationship to Relations: One-One (Approach C)



R1: OfficialTrailer(Url, Content, RunningTime)

R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

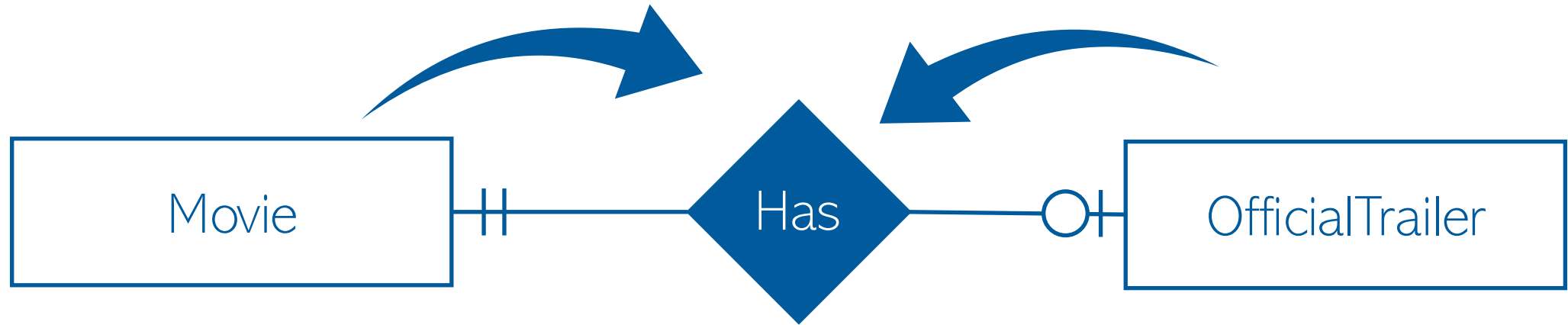
R3: MovieOfficialTrailer (Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay, OfficialTrailer.Url)

FK₁: Optional or Mandatory?

FK₂: Optional or Mandatory?

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Relationship to Relations: One-One (Approach C)



R1: OfficialTrailer(Url, Content, RunningTime)

R2: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R3: MovieOfficialTrailer (Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay, OfficialTrailer.Url)

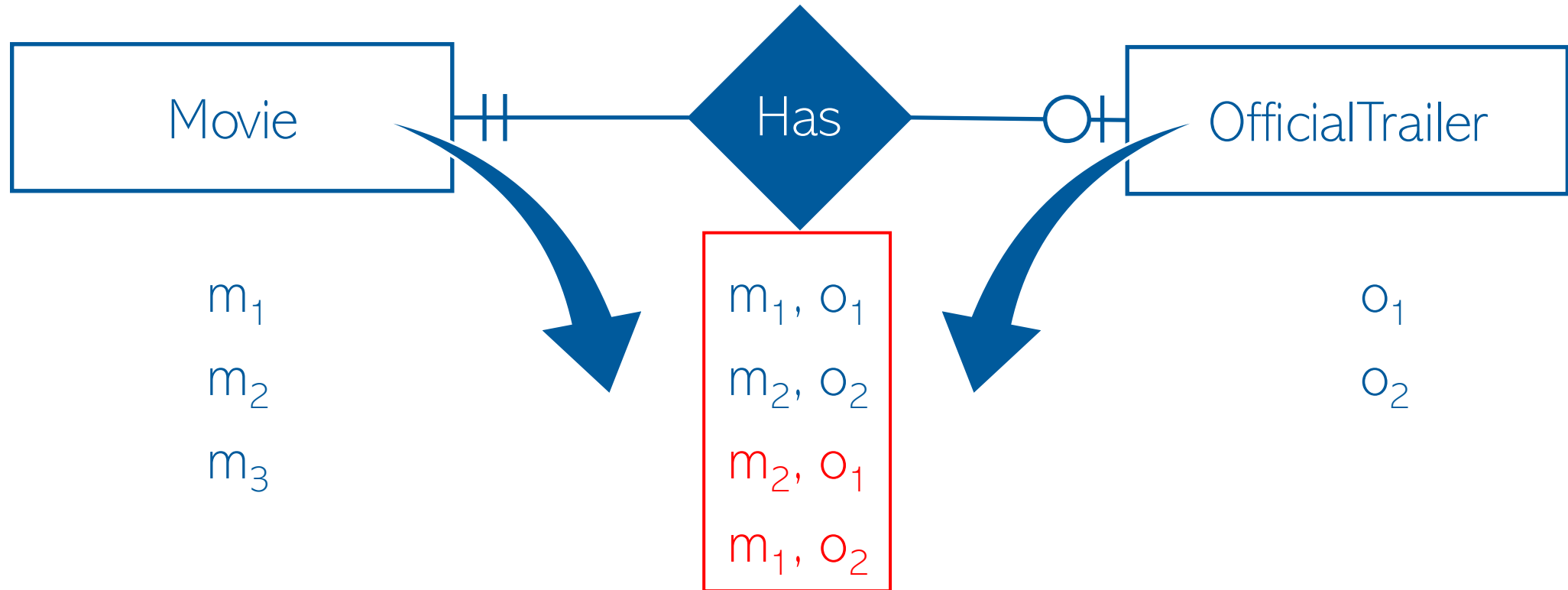
FK₁: Mandatory

FK₂: Mandatory

If these relationships exist, it means that they must have values.

79

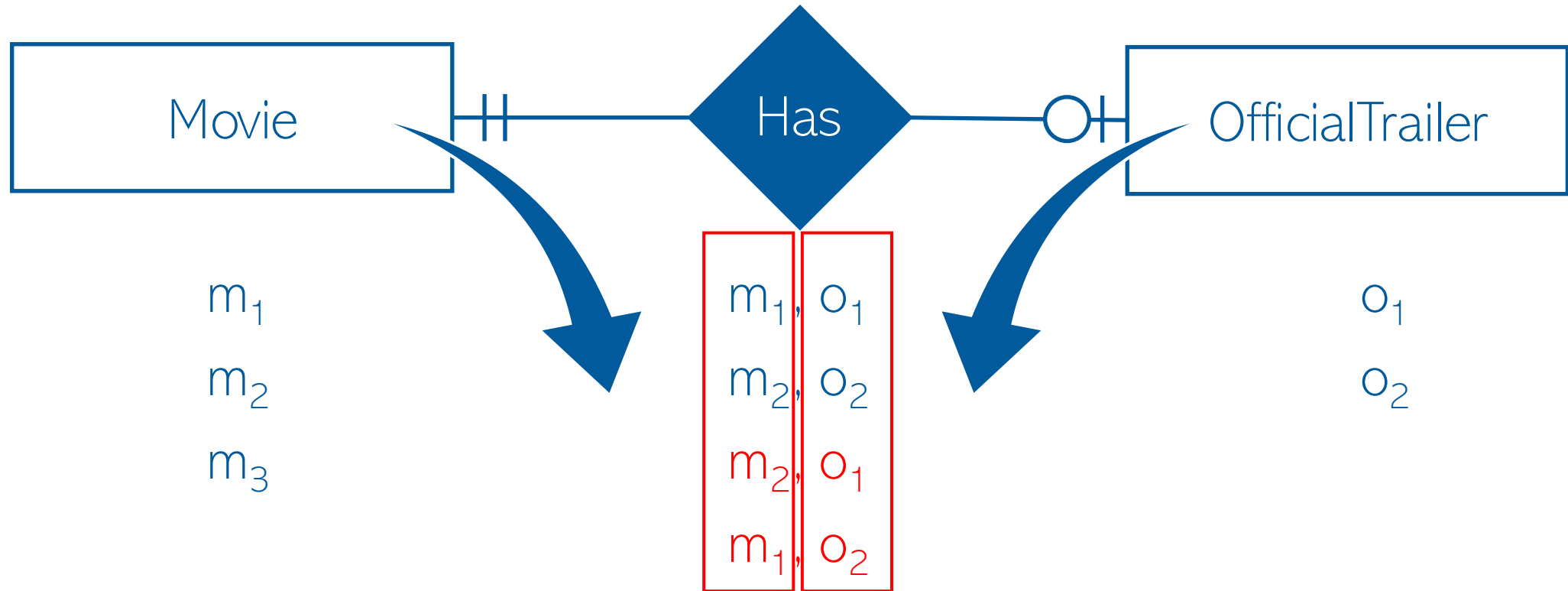
Relationship to Relations: One-One (Approach C)



Primary Key does not prevent these tuples ...

80

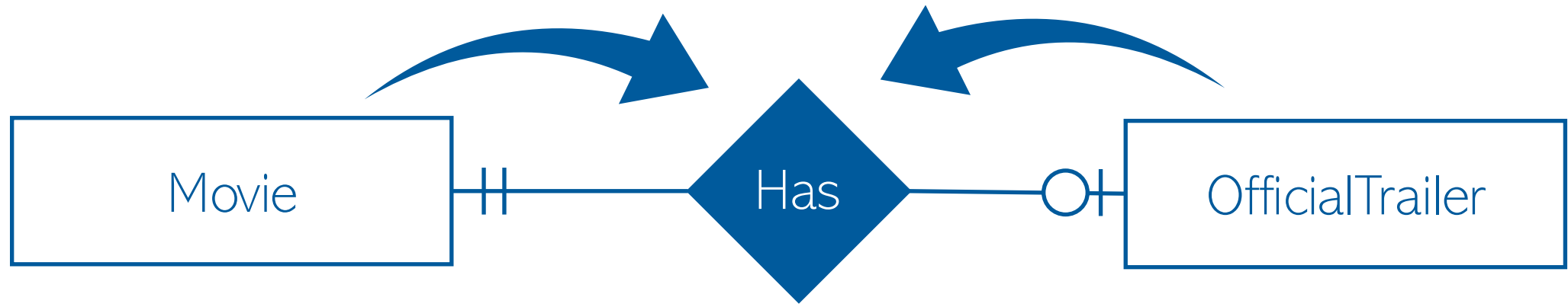
Relationship to Relations: One-One (Approach C)



Each domain (column) must be unique ...

81

Relationship to Relations: One-One (Approach C)



R1: OfficialTrailer(Url, Content, RunningTime)

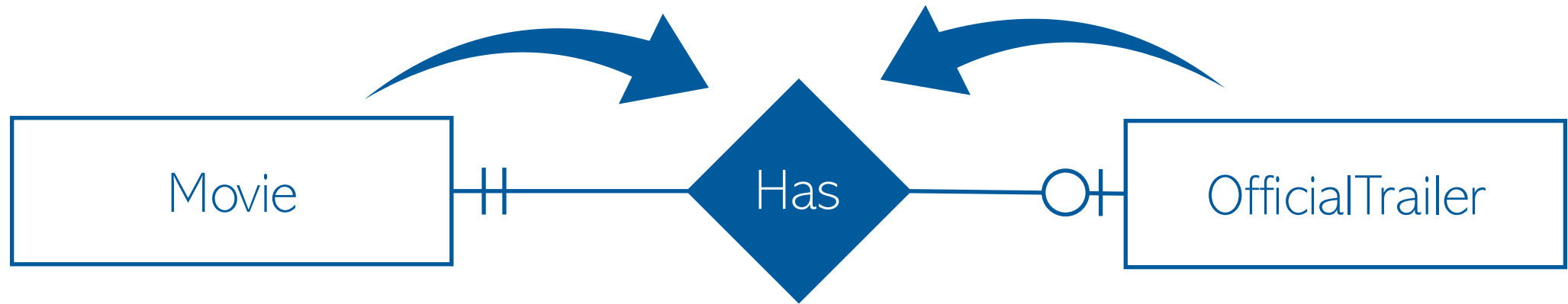
R2: Movie(Title, RunningTime, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R3: MovieOfficialTrailer (Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay, OfficialTrailer.Url)

CK={OfficialTrailer.Url} and Mandatory

82

Relationship to Relations: One-One (Approach C)



R1: OfficialTrailer(Url, Content, RunningTime)

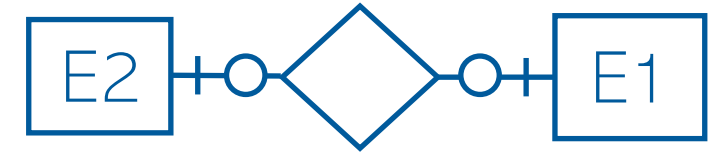
R2: Movie(Title, RunningTime, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R3: MovieOfficialTrailer(Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay, OfficialTrailer.Url)

CK={Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay} and Mandatory

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Relationship to Relations: One-One



Input: One-One relationship btw. E2 and E1, i.e.,

Output: Relations R1 for E1, R2 for E2, and/or R3 for relationship.

- 1) For E1, create relation R1 with the same attributes and keys as in E1
- 2) For E2, create relation R2 with the same attributes and keys as in E2
- 3) Do only one of the followings
 - a) [Foreign & Candidate Key set FK, CK] Add key set of E1 to R2 & make them candidate key of R2
If E1's ordinality is mandatory, make CK mandatory
 - b) [Foreign & Candidate Key Set FK, CK] Add key set of E2 to R1 & make them candidate key of R1
If E2's ordinality is mandatory, make CK mandatory
 - c) Create new relation R3
 - I) [Foreign & Primary Key set FK, PK] Add key set of E1 to R3 & make them primary key of R3
 - II) [Foreign & Candidate Key set FK, CK] Add key set of E2 to R3 & make them candidate key of R3 and mandatory



Divine Intervention ..., Pulp Fiction, Quintine Tarantino, 1994
<https://www.youtube.com/watch?v=l2oaQ3Qv7pQ>

85

Surrogate Key vs. Natural (Business) Key Sets

Sometimes Primary Key set gets large,

E.g., Movie's PK={Title, ReleaseYear, ReleaseMonth, ReleaseDay}

E.g., Actor's PK={FirstName, LastName, BirthDate, BirthPlace}

It replicates when becomes FK to other relations.

Not convenient!

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Surrogate Key vs. Natural (Business) Key Sets

- One single domain (attribute) key (not a set)
- System-generated, e.g., AUTONUMBER, i.e., 1,2,3,....,
- The value has no meaning in real-world!
- Just for data engineering convenience.
- Usually named Id, ID, RowID, RowId, TupleId, ...

Old natural (business) primary key becomes mandatory candidate key.

Old natural (business) primary key becomes mandatory candidate key.

Old natural (business) primary key becomes mandatory candidate key.

Old natural (business) primary key becomes mandatory candidate key.

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Surrogate Key vs. Natural (Business) Key Sets

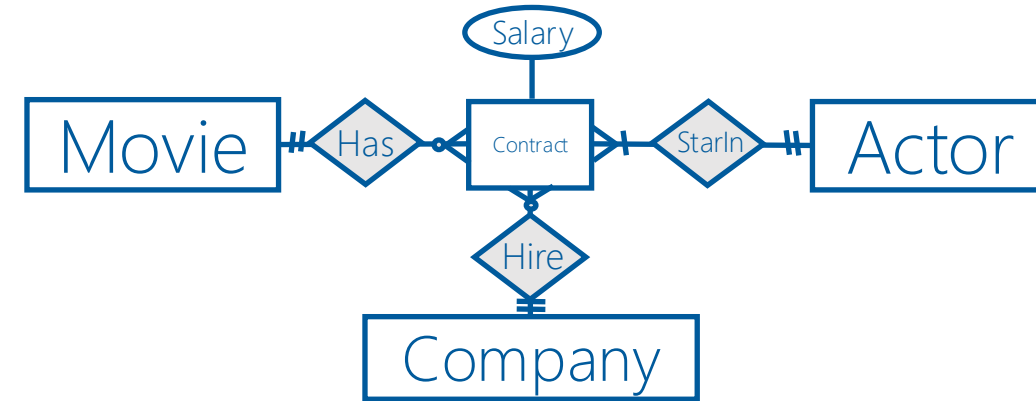
Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, ...)



Movie(Id, Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, ...)
CK={Title, ReleaseYear, ReleaseMonth, ReleaseDay} and Mandatory

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Surrogate Key vs. Natural (Business) Key Sets



R1: Company(Id, Name, Address, Phone, ...)

CK={Name} Mandatory

R2: Actor(Id, FirstName, LastName, BirthDate, BirthPlace, ...)

CK={FirstName, LastName, BirthDate, BirthPlace} Mandatory

R3: Movie(Id, Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime)

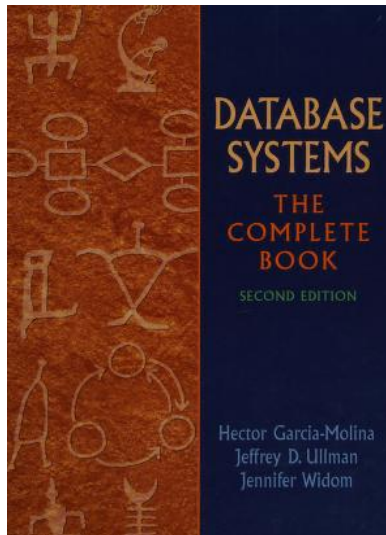
CK={Title, ReleaseYear, ReleaseMonth, ReleaseDay} Mandatory

R4: Contract(Id, Salary, Company.Id, Actor.Id, Movie.Id)

FK₁={Company.Id}, FK₂={Actor.Id}, FK₃={Movie.Id}

CK=FK₁ U FK₂ U FK₃ Mandatory

Outline: W05A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation (One-One)

III) IS-A to Relation

Hands-On



Trunk POV..., Pulp Fiction, Quintine Tarantino, 1994
<https://www.youtube.com/watch?v=edFrWONapIE>

ISA to Relation